

Project Data Page:

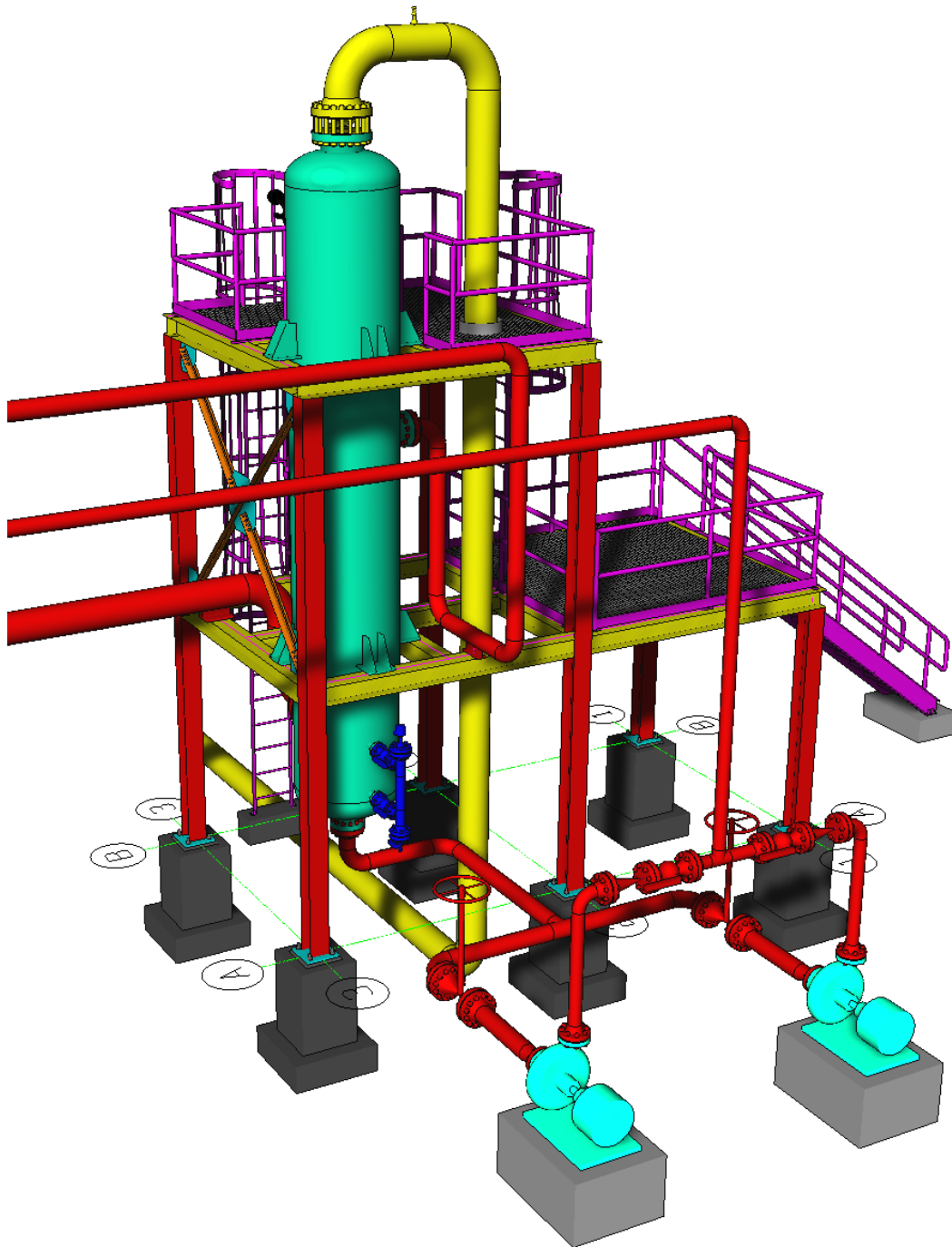




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : ...\\406-V-098R07%.pvdb

Date of Analysis : Jul 23, 2026 2:29pm

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2060.000 mm ID
 Vessel Design Length, Tangent to Tangent 6180.00 mm
 Specified Datum Line Distance 0.00 mm
 Internal Design Temperature 65 °C
 Internal Design Pressure 0.380 MPa
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 0 MPa
 Shop Test Pressure 0.558 MPa
 Required Minimum Design Metal Temperature 11.4 °C
 Warmest Computed Minimum Design Metal Temperature -37.0 °C
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Left D end	0.399	0.00	0.000	3.0000	No	No
Shell 1	0.399	0.00	1.000	3.0000	N/A	No
Shell 2	0.399	0.00	1.000	3.0000	N/A	No
Shell 3	0.399	0.00	1.000	3.0000	N/A	No
Right D end	0.399	0.00	0.000	3.0000	No	No

Liquid Level: 1915.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.993

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Torisph	50.0	50.0	12.0	10.0	9.2	...	0.85	0.85
Cylinder	2550.0	2500.0	10.0	10.0	6.5	...	0.85	0.85
Cylinder	3630.0	1080.0	10.0	10.0	6.5	...	0.85	0.85
Cylinder	6130.0	2500.0	10.0	10.0	6.5	...	0.85	0.85
Torisph	6180.0	50.0	12.0	10.0	9.2	...	0.85	0.85

Local Stress Analysis Results:



Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_6"	WRC-297	0.998	Shell	Passed
N4_4"	WRC-297	0.897	Nozzle	Passed
N1_10"	WRC-297	1.295	Shell	Failed
N2_4"	WRC-297	0.551	Nozzle	Passed
N5_4"	WRC-297	0.993	Nozzle	Passed
N6_4"	WRC-297	0.822	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	250.000	mm
Saddle Bearing Angle	148.000	deg.
Centerline Dimension	1230.000	mm
Wear Pad Width	350.000	mm
Wear Pad Thickness	10.000	mm
Wear Pad Bearing Angle	160.000	deg.
Distance from Saddle to Tangent	750.000	mm

Baseplate Length	2030.000	mm
Baseplate Thickness	19.000	mm
Baseplate Width	300.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	10.000	mm
Web Thickness	10.000	mm
Height of Center Web	161.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	188288.77	N
Maximum Transverse Saddle Shear Load	22063.40	N
Maximum Longitudinal Saddle Shear Load	34973.93	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	159369.94	N
Maximum Transverse Saddle Shear Load	7280.92	N
Maximum Longitudinal Saddle Shear Load	5029.39	N

Weights:

Fabricated - Bare w/o removable internals	5303.3	kgm
Shop Test - Fabricated + water (full)	27304.3	kgm
Shipping - Fab. + rem. intls.+ shipping app.	7772.4	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	7772.4	kgm
Empty - Fab. + intls. + details + wghts.	7772.4	kgm
Operating - Empty + operating liquid (no ca)	28975.6	kgm

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
M1_8"	...	...	...	...		OK	Passed
K2A_2"	1	...	...	OK	5.85	OK	No Calc[*]
N3_6"	1	...	...	OK	5.85	OK	Passed
N4_4"	1	...	...	OK	5.85	OK	Passed
K2B_2"	1	...	...	OK	5.99	OK	No Calc[*]
K3_2"	1	...	...	OK	5.85	OK	No Calc[*]
K4_2"	1	...	...	OK	5.85	OK	No Calc[*]
K1A_2"	1	...	...	OK	5.85	OK	No Calc[*]
K1B_2"	1	...	...	OK	5.99	OK	No Calc[*]
N1_10"	1	...	...	OK	5.99	OK	Passed
N2_4"	3	...	...	OK	4.00	OK	Passed
N5_4"	1	...	...	OK	5.85	OK	Passed
N6_4"	1	...	...	OK	5.85	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 0 Nozzle : M18"
 Minimum MAWP Shells/Flanges : 0 Element : Right D end
 Minimum MAPnc Shells/Flanges : 1 Element : Right D end

Computed Vessel M.A.W.P. : 0 MPa

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	M1_8"	0.000	0.000	415.524
20	K2A_2"	450.000	90.000	107.600
20	N3_6"	1050.000	90.000	304.145
20	N4_4"	1850.000	90.000	204.641
20	K2B_2"	450.000	270.000	107.600
30	K3_2"	2850.000	90.000	107.600
30	K4_2"	3150.000	90.000	107.600
40	K1A_2"	5330.000	90.000	107.600
40	K1B_2"	5830.000	270.000	107.600
40	N1_10"	4330.000	270.000	499.293
40	N2_4"	0.000	270.000	213.531
40	N5_4"	4330.000	90.000	204.641
40	N6_4"	4930.000	90.000	195.662

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K2A_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	290.73	150
K2B_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	240.73	150
K3_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	290.73	150
K4_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	290.73	150
K1A_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	290.73	150
K1B_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	220.28	150
N4_4"	4.000 in	80	WNF	4.500	8.560	214.30	10.00	241.59	300
N2_4"	4.000 in	40	WNF	4.500	6.020	216.00	16.00	277.24	300
N5_4"	4.000 in	80	WNF	4.500	8.560	203.20	6.00	241.59	300
N6_4"	4.000 in	120	WNF	4.500	11.125	203.20	6.00	241.59	300
N3_6"	6.000 in	80	WNF	6.625	10.973	328.27	10.00	243.44	150
M1_8"	8.000 in	40	WNF	8.625	8.179	...	...	162.88	150
N1_10"	10.000 in	80	WNF	10.750	15.088	573.05	10.00	269.09	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K2A_2"	SA-105	6.000	12.000	...	...	...
K2B_2"	SA-105	6.000	12.000	...	...	...
K3_2"	SA-105	6.000	12.000	...	...	...
K4_2"	SA-105	6.000	12.000	...	...	...
K1A_2"	SA-105	6.000	12.000	...	...	...
K1B_2"	SA-105	6.000	12.000	...	...	...
N4_4"	SA-106 B	6.000	6.000	10.000	10.000	...
N2_4"	SA-106 B	16.000	6.000	12.000	16.000	...
N5_4"	SA-106 B	6.000	6.000	6.000	6.000	...
N6_4"	SA-106 B	6.000	6.000	6.000	6.000	...
N3_6"	SA-106 B	6.000	10.000	10.000	10.000	...
M1_8"	SA-106 B	8.000	10.000	...	...	...
N1_10"	SA-106 B	6.000	10.000	10.000	10.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K2A_2"	450.000	90.0	280.00	0.00	Shell 1
K2B_2"	450.000	270.0	230.00	0.00	Shell 1
K3_2"	2850.000	90.0	280.00	0.00	Shell 2
K4_2"	3150.000	90.0	280.00	0.00	Shell 2
K1A_2"	5330.000	90.0	280.00	0.00	Shell 3
K1B_2"	5830.000	270.0	209.55	0.00	Shell 3
N4_4"	1850.000	90.0	230.00	0.00	Shell 1
N2_4"	3730.000	270.0	250.00	0.00	N1_10"
N5_4"	4330.000	90.0	230.00	0.00	Shell 3
N6_4"	4930.000	90.0	230.00	0.00	Shell 3
N3_6"	1050.000	90.0	230.00	0.00	Shell 1

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FileName : 406-V-098_R0_7% Page 6 of 170
Nozzle Schedule: Step: 32 2:29pm Jul 23,2026



M1_8"	...	0.0	150.00	0.00	Left D end
N1_10"	4330.000	270.0	250.00	0.00	Shell 3

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	TORISPHERICAL HEAD: 12.0mm THK X 2060.0mm ID X 50.0mm	SA-516 70
2	SADDLE: 250.0mm X 148 DEG	SA-516 70
1	PLATFORM: 1200mm X 2072mm WIDE	
1	LADDER: 3502.0mm	...
1	CLASS 150 GR 1.1, 8.0" BLIND FLANGE(S)	SA-105
4	CYLINDRICAL SEGMENT 10.0mm THK X 3235.8mm X 2500.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 10.0mm THK X 3235.8mm X 1080.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Left D end	[10]	B	-29	-37	-29	0.849	10.000	0.85	No
Left D end	[7]	B	-23	-48	-29	0.374	12.000	0.85	No
Shell 1	[8]	B	-29	-48	-29	0.480	10.000	0.85	No
Shell 2	[8]	B	-29	-48	-29	0.480	10.000	0.85	No
Shell 3	[8]	B	-29	-48	-29	0.480	10.000	0.85	No
Right D end	[10]	B	-29	-37	-29	0.850	10.000	0.85	No
Right D end	[7]	B	-23	-48	-29	0.374	12.000	0.85	No
Fixed Saddle	[24]	B	-29	-48	-29	0.480	10.000	0.85	No
New Saddle	[24]	B	-29	-48	-29	0.480	10.000	0.85	No
M18"	[1]	B	-8	-104	-29	0.064	7.156	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2A2"	[1]	A	-8	-48	-29	0.407	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N36"	[1]	B	-29	-48	-29	0.407	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N44"	[1]	B	-29	-48	-29	0.407	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2B2"	[1]	A	-8	-48	-29	0.427	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K32"	[1]	A	-8	-48	-29	0.407	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K42"	[1]	A	-8	-48	-29	0.407	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1A2"	[1]	A	-8	-48	-29	0.407	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1B2"	[1]	A	-8	-48	-29	0.427	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N110"	[1]	B	-29	-48	-29	0.427	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N24"	[1]	B	-8	-104	-29	0.051	5.267	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N54"	[1]	B	-29	-48	-29	0.407	6.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N64"	[1]	B	-29	-48	-29	0.407	6.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					

Warmest MDMT: -8 -37

Required Minimum Design Metal Temperature 11.4 °C
 Warmest Computed Minimum Design Metal Temperature -37.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld



[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	0.38	MPa
Design Internal Temperature	65.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	11.4	°C
Type of Construction	Welded	
Special Service	Air/Water/Steam	
Degree of Radiography	RT-2	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050



1.600 0.044
 1.800 0.039
 2.000 0.035
 2.400 0.029
 0.000 0.000

Consider Vertical Acceleration Yes
 Design Pressure + Static Head Y
 Consider MAP New and Cold in Noz. Design N
 Consider External Loads for Nozzle Des. Y
 Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element# 1/5
 Element From Node 10
 Element To Node 20
 Element Type Torispherical
 Description Left D end
 Distance "FROM" to "TO" 50 mm
 Inside Diameter 2060 mm
 Element Thickness 10 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 12 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 0.38 MPa
 Design Temperature Internal Pressure 65 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Allowable Stress, Ambient 138 MPa
 Allowable Stress, Operating 138 MPa
 Allowable Stress, Hydrotest 234 MPa
 Material Density 0.00775 kg/cm³
 P Number Thickness 32.004 mm
 Yield Stress, Operating 246 MPa
 UCS-66 Chart Curve Designation B
 External Pressure Chart Name CS-2
 UNS Number K02700
 Product Form Plate
 Efficiency, Longitudinal Seam 0.85
 Efficiency, Circumferential Seam 0.85
 Tori Head Crown Radius 2077 mm
 Tori Head Knuckle Radius 124.62 mm
 Weld is pre-Heated No
 Element From Node 10
 Detail Type Liquid



Detail ID	Liquid 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1915	mm
Liquid Density	0.0009917	kg/cm³

Element From Node	10	
Detail Type	Nozzle	
Detail ID	M18"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	8	in.
Nozzle Schedule	40	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	10	
Detail Type	Weight	
Detail ID	Handle Wt	
Dist. from "FROM" Node / Offset dist	-347	mm
Miscellaneous Weight	4.903	N
Offset from Element Centerline	0	mm

Element From Node	10	
Detail Type	Weight	
Detail ID	Manway davit	
Dist. from "FROM" Node / Offset dist	50	mm
Miscellaneous Weight	4.903	N
Offset from Element Centerline	0	mm

Element#	2/5	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	Shell 1	
Distance "FROM" to "TO"	2500	mm
Inside Diameter	2060	mm
Element Thickness	10	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	10	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.38	MPa
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Saddle	
Detail ID	Fixed Saddle	
Dist. from "FROM" Node / Offset dist	700	mm
Width of Saddle	250	mm
Height of Saddle at Bottom	1230	mm
Saddle Contact Angle	148.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	350	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	160.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	



Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	20	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	9123860	mm ²
Platform Weight	22628	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	2072	mm
Platform Height	1200	mm
Platform Clearance or End Offset	230	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	-500	mm
Ladder End Elevation	3002	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	6180	mm
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1915	mm
Liquid Density	0.0009916	kg/cm ³
Element From Node	20	
Detail Type	Nozzle	
Detail ID	K2A2"	
Dist. from "FROM" Node / Offset dist	400	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N36"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N44"	
Dist. from "FROM" Node / Offset dist	1800	mm
Nozzle Diameter	4	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	20	



Detail Type	Nozzle
Detail ID	K2B2"
Dist. from "FROM" Node / Offset dist	400 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	20
Detail Type	Weight
Detail ID	Lifting Wt 1
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	392.24 N
Offset from Element Centerline	1000 mm

Element From Node	20
Detail Type	Weight
Detail ID	Lifting wt 2
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	392.24 N
Offset from Element Centerline	1000 mm

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Shell 2
Distance "FROM" to "TO"	1080 mm
Inside Diameter	2060 mm
Element Thickness	10 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.38 MPa
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1915 mm
Liquid Density	0.0009917 kg/cm³

Element From Node	30
Detail Type	Nozzle
Detail ID	K32"
Dist. from "FROM" Node / Offset dist	300 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	30
-------------------	----



Detail Type	Nozzle
Detail ID	K42"
Dist. from "FROM" Node / Offset dist	600 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	30
Detail Type	Weight
Detail ID	Clip WT
Dist. from "FROM" Node / Offset dist	150 mm
Miscellaneous Weight	784.48 N
Offset from Element Centerline	0 mm

Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell 3
Distance "FROM" to "TO"	2500 mm
Inside Diameter	2060 mm
Element Thickness	10 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.38 MPa
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	1800 mm
Width of Saddle	250 mm
Height of Saddle at Bottom	1230 mm
Saddle Contact Angle	148.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	350 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	160.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1915 mm
Liquid Density	0.0009916 kg/cm³

Element From Node	40
Detail Type	Nozzle
Detail ID	K1A2"



Dist. from "FROM" Node / Offset dist	1700	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K1B2"	
Dist. from "FROM" Node / Offset dist	2200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N110"	
Dist. from "FROM" Node / Offset dist	700	mm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N24"	
Dist. from "FROM" Node / Offset dist	100	mm
Nozzle Diameter	4	in.
Nozzle Schedule	40	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N54"	
Dist. from "FROM" Node / Offset dist	700	mm
Nozzle Diameter	4	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N64"	
Dist. from "FROM" Node / Offset dist	1300	mm
Nozzle Diameter	4	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	90.0	



Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	40
Detail Type	Weight
Detail ID	PSV
Dist. from "FROM" Node / Offset dist	2000 mm
Miscellaneous Weight	0 N
Offset from Element Centerline	1300 mm
Element From Node	40
Detail Type	Weight
Detail ID	Lifting wt 3
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	392.24 N
Offset from Element Centerline	1000 mm
Element From Node	40
Detail Type	Weight
Detail ID	Lifting wt 4
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	392.24 N
Offset from Element Centerline	1000 mm

Element#	5/5
Element From Node	50
Element To Node	60
Element Type	Torispherical
Description	Right D end
Distance "FROM" to "TO"	50 mm
Inside Diameter	2060 mm
Element Thickness	10 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.38 MPa
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Tori Head Crown Radius	2078.5 mm
Tori Head Knuckle Radius	124.71 mm
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1915 mm
Liquid Density	0.0009917 kg/cm³

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Left D end		50	...	50	...
Shell 1		2550	...	2500	...
Shell 2		3630	...	1080	...
Shell 3		6130	...	2500	...
Right D end		6180	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Left D end		0.3987	12	3	2060	117.3	1
Shell 1		0.3987	10	3	2060	117.3	2
Shell 2		0.3987	10	3	2060	117.3	3
Shell 3		0.3987	10	3	2060	117.3	4
Right D end		0.3987	12	3	2060	117.3	5

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Left D end		0.38002	0.42992	0.63756	10	9.2211	1
Shell 1		0.38002	0.77296	1.13222	10	6.51833	2
Shell 2		0.38002	0.77296	1.13222	10	6.51834	3
Shell 3		0.38002	0.77296	1.13222	10	6.51833	4
Right D end		0.38002	0.42959	0.6371	10	9.22561	5

Minimum 0 1

MAWP: 0 MPa, limited by: Right D end.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Left D end

Torispherical Head Design Information:					
Design Pressure	P	0.399 MPa	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2060.000 mm	Actual Stress		122.6 MPa
Spec. Min. Thk t,e, or sa		10.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		4403763.0 mm²	SG of Contents		0.993
Element Volume		873.4 ltr	Weight of Contents		8381.1 N
Empty Weight		3956.3 N	Operating Weight		12337.4 N
			Comp. All. Ope. Stress		95.8 MPa
Ins. Crn. Rad. L or R		2077.0000 mm	Ins. Knuckle Radius	r	124.620 mm

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.875 mm

Inside Corroded Head Depth [h]:

$$= L - \sqrt{((L - Di / 2) \cdot (L + Di / 2 - 2 \cdot r))}$$

$$= 2080.0 - \sqrt{((2080.0 - 2066.0 / 2) \cdot (2080.0 + 2066.0 / 2 - 2 \cdot 127.62))}$$

$$= 350.239 \text{ mm}$$

M factor for Torispherical Heads (Corroded):

$$= (3 + \sqrt{((L + C) / (r + C))}) / 4 \text{ per Appendix 1-4 (b \& d)}$$

$$= (3 + \sqrt{((2077.0 + 3.0) / (124.62 + 3.0))}) / 4$$

$$= 1.7593$$

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot L \cdot M) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4 (d)}$$

$$= (0.399 \cdot 2080.0 \cdot 1.7593) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 0.399)$$

$$= 6.2211 + 3.0000 = 9.2211 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

[Less Operating Hydrostatic Head Pressure of 0.019 MPa](#)

$$= (2 \cdot S \cdot E \cdot t) / (M \cdot L + 0.2 \cdot t) \text{ per Appendix 1-4 (d)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 7.0) / (1.7593 \cdot 2080.0 + 0.2 \cdot 7.0)$$

$$= 0 - 0 = 0 \text{ MPa}$$

M factor for Torispherical Heads (New & Cold):

$$= (3 + \sqrt{(L / r)}) / 4 \text{ per Appendix 1-4 (b \& d)}$$

$$= (3 + \sqrt{(2077.0 / 124.62)}) / 4$$

$$= 1.7706$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (M \cdot L + 0.2 \cdot t) \text{ per Appendix 1-4 (d)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 10.0) / (1.7706 \cdot 2077.0 + 0.2 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (M \cdot L + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$



$$= (0.399 \cdot (1.7593 \cdot 2080.0 + 0.2 \cdot 7.0)) / (2 \cdot 0.85 \cdot 7.0)$$

$$= 122.648 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (0.399 \cdot 1033.0) / (138.0 \cdot 0.85 - 0.6 \cdot 0.399) + 3.0$$

$$= 6.518 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.019 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 9.0) / (1033.0 + 0.6 \cdot 9.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.890 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10.0$, $t_r = 6.995$, $c = 3.0 \text{ mm}$, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.849$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12.0$, $t_r = 3.957$, $c = 3.0 \text{ mm}$, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.374$, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.448 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.849	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-37 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.875 mm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	0.399 MPa	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2500.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2060.000 mm	Actual Stress		69.5 MPa
Spec. Min. Thk t,e, or sa		10.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		16336282.0 mm²	SG of Contents		0.993
Element Volume		8333.8 ltr	Weight of Contents		78598.4 N
Empty Weight		12356.0 N	Operating Weight		90954.4 N
			Comp. All. Ope. Stress		95.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.399 \cdot 1033.0) / (138.0 \cdot 0.85 - 0.6 \cdot 0.399)$$

$$= 3.5183 + 3.0000 = 6.5183 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.019 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 7.0) / (1033.0 + 0.6 \cdot 7.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 10.0) / (1030.0 + 0.6 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.399 \cdot (1033.0 + 0.6 \cdot 7.0)) / (0.85 \cdot 7.0)$$

$$= 69.500 \text{ MPa}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 0.483 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 10.0, tr = 3.957, c = 3.0 mm, E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.48, Temp. Reduction = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.448 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.480	Temperature Difference	19 °C

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Element: Shell 1 Step: 5 2:29pm Jul 23,2026



HEXAGON

UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	0.399 MPa	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		1080.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2060.000 mm	Actual Stress	69.5 MPa	
Spec. Min. Thk t _e , or s _a		10.0000 mm	Spec. Nominal Thk.	10.000 mm	
Surface Area		7057274.0 mm²	SG of Contents	0.993	
Element Volume		3600.2 ltr	Weight of Contents	33956.2 N	
Empty Weight		5337.8 N	Operating Weight	39294.0 N	
			Comp. All. Ope. Stress	95.8 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.399 \cdot 1033.0) / (138.0 \cdot 0.85 - 0.6 \cdot 0.399)$$

$$= 3.5183 + 3.0000 = 6.5183 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.019 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 7.0) / (1033.0 + 0.6 \cdot 7.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 10.0) / (1030.0 + 0.6 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.399 \cdot (1033.0 + 0.6 \cdot 7.0)) / (0.85 \cdot 7.0)$$

$$= 69.500 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.483 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 10.0, t_r = 3.957, c = 3.0 mm, E* = 0.85
 Thickness Ratio = t_r * E* / (t_g - c) = 0.48, Temp. Reduction = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.448 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.480	Temperature Difference	19 °C

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Element: Shell 2 Step: 6 2:29pm Jul 23,2026



UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell 3

Cylindrical Shell Design Information:					
Design Pressure	P	0.399 MPa	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2500.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2060.000 mm	Actual Stress		69.5 MPa
Spec. Min. Thk t,e, or sa		10.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		16336282.0 mm²	SG of Contents		0.993
Element Volume		8333.8 ltr	Weight of Contents		78598.4 N
Empty Weight		12356.0 N	Operating Weight		90954.4 N
			Comp. All. Ope. Stress		95.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.399 \cdot 1033.0) / (138.0 \cdot 0.85 - 0.6 \cdot 0.399)$$

$$= 3.5183 + 3.0000 = 6.5183 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.019 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 7.0) / (1033.0 + 0.6 \cdot 7.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 10.0) / (1030.0 + 0.6 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.399 \cdot (1033.0 + 0.6 \cdot 7.0)) / (0.85 \cdot 7.0)$$

$$= 69.500 \text{ MPa}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 0.483 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 10.0, tr = 3.957, c = 3.0 mm, E* = 0.85
 Thickness Ratio = tr * E*/(tg - c) = 0.48, Temp. Reduction = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.448 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.480	Temperature Difference	19 °C

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Element: Shell 3 Step: 7 2:29pm Jul 23,2026



UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Right D end

Torispherical Head Design Information:					
Design Pressure	P	0.399 MPa	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	2060.000 mm	Actual Stress		122.7 MPa
Spec. Min. Thk t,e, or sa		10.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		4403756.0 mm²	SG of Contents		0.993
Element Volume		873.3 ltr	Weight of Contents		8379.9 N
Empty Weight		3956.2 N	Operating Weight		12336.2 N
			Comp. All. Ope. Stress		95.8 MPa
Ins. Crn. Rad. L or R		2078.5000 mm	Ins. Knuckle Radius	r	124.710 mm

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.875 mm

Inside Corroded Head Depth [h]:

$$= L - \sqrt{((L - D_i / 2) \cdot (L + D_i / 2 - 2 \cdot r))}$$

$$= 2081.5 - \sqrt{((2081.5 - 2066.0 / 2) \cdot (2081.5 + 2066.0 / 2 - 2 \cdot 127.71))}$$

$$= 350.1 \text{ mm}$$

M factor for Torispherical Heads (Corroded):

$$= (3 + \sqrt{((L + C) / (r + C))}) / 4 \text{ per Appendix 1-4 (b \& d)}$$

$$= (3 + \sqrt{((2078.5 + 3.0) / (124.71 + 3.0))}) / 4$$

$$= 1.7593$$

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot L \cdot M) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4 (d)}$$

$$= (0.399 \cdot 2081.5 \cdot 1.7593) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 0.399)$$

$$= 6.2256 + 3.0000 = 9.2256 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

[Less Operating Hydrostatic Head Pressure of 0.019 MPa](#)

$$= (2 \cdot S \cdot E \cdot t) / (M \cdot L + 0.2 \cdot t) \text{ per Appendix 1-4 (d)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 7.0) / (1.7593 \cdot 2081.5 + 0.2 \cdot 7.0)$$

$$= 0 - 0 = 0 \text{ MPa}$$

M factor for Torispherical Heads (New & Cold):

$$= (3 + \sqrt{(L / r)}) / 4 \text{ per Appendix 1-4 (b \& d)}$$

$$= (3 + \sqrt{(2078.5 / 124.71)}) / 4$$

$$= 1.7706$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (M \cdot L + 0.2 \cdot t) \text{ per Appendix 1-4 (d)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 10.0) / (1.7706 \cdot 2078.5 + 0.2 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (M \cdot L + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.399 \cdot (1.7593 \cdot 2081.5 + 0.2 \cdot 7.0)) / (2 \cdot 0.85 \cdot 7.0)$$

$$= 122.736 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (0.399 \cdot 1033.0) / (138.0 \cdot 0.85 - 0.6 \cdot 0.399) + 3.0$$

$$= 6.518 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.019 MPa](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 9.0) / (1033.0 + 0.6 \cdot 9.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.885 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10.0$, $t_r = 7.0$, $c = 3.0$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12.0$, $t_r = 3.957$, $c = 3.0$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.374$, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.448 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-37 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 25.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 12.875 mm

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.558 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.494 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.808 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.473 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.543 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.543 MPa
App 27-4	= M.A.W.P.	0.430 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 0.43 • 1.0
 = 0.558 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Left D end	125.2	234.0	0.535	0.58
Shell 1	70.5	234.0	0.301	0.58
Shell 2	70.5	234.0	0.301	0.58
Shell 3	70.5	234.0	0.301	0.58
Right D end	125.3	234.0	0.536	0.58

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1_8"	Nozzle	118.00	118.00	1.000
K2A_2"	Nozzle	138.00	138.00	1.000
N3_6"	Nozzle	118.00	118.00	1.000
N3_6"	Pad	138.00	138.00	1.000
N4_4"	Nozzle	118.00	118.00	1.000
N4_4"	Pad	138.00	138.00	1.000
K2B_2"	Nozzle	138.00	138.00	1.000
K3_2"	Nozzle	138.00	138.00	1.000
K4_2"	Nozzle	138.00	138.00	1.000
K1A_2"	Nozzle	138.00	138.00	1.000
K1B_2"	Nozzle	138.00	138.00	1.000
N1_10"	Nozzle	118.00	118.00	1.000
N1_10"	Pad	138.00	138.00	1.000
N2_4"	Nozzle	118.00	118.00	1.000
N2_4"	Pad	138.00	138.00	1.000
N5_4"	Nozzle	118.00	118.00	1.000
N5_4"	Pad	138.00	138.00	1.000
N6_4"	Nozzle	118.00	118.00	1.000
N6_4"	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Left D end	138.00	138.00	1.000
Shell 1	138.00	138.00	1.000
Shell 2	138.00	138.00	1.000
Shell 3	138.00	138.00	1.000
Right D end	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
M1_8"	8.63	216.00	0.040
K2A_2"	1.44	225.00	0.006
N3_6"	4.84	216.00	0.022
N4_4"	4.18	216.00	0.019
K2B_2"	1.44	225.00	0.006
K3_2"	1.44	225.00	0.006
K4_2"	1.44	225.00	0.006
K1A_2"	1.44	225.00	0.006
K1B_2"	1.44	225.00	0.006
N1_10"	5.75	216.00	0.027
N2_4"	6.05	216.00	0.028
N5_4"	4.18	216.00	0.019
N6_4"	3.17	216.00	0.015

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	2087	7	0.00041926	41.9168
20	30	6411.45	2080	7	0.80262E-04	8.02437
30	40	6411.45	2080	7	0.80262E-04	8.02437
40	50	6411.45	2080	7	0.80262E-04	8.02437
50	60	No Calc	2088.5	7	0.00041896	41.8867

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	10	No Calc	...	0.14059
20	30	10	No Calc	...	0.036007
30	40	10	No Calc	...	0.036007
40	50	10	No Calc	...	0.036007
50	60	10	No Calc	...	0.14039

Minimum

0

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	6411.45	No Calc	No Calc	No Calc
30	40	6411.45	No Calc	No Calc	No Calc
40	50	6411.45	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	403.458	873.41	303.735	886.279	28.242
20	30	1260.05	8333.78	883.313	8382.4	88.2035
30	40	544.342	3600.19	381.592	3621.2	38.104
40	50	1260.05	8333.78	883.313	8382.4	88.2035
50	60	403.452	873.285	303.735	886.153	28.2416
Total		3871	22014.46	2755	22158.43	270

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	854.694	-115.746	72.5	...	Liquid 10
10	Nozl	47.0539	-347.238	...	...	M1_8"
10	Wght	0.5	-347.238	...	...	Handle Wt
10	Wght	0.5	-347.238	...	...	Manway davit
20	Sadl	319.296	700	1120	...	Fixed Saddle
20	Plat	2469.13	3090	1116.2	...	PLAT:[1 OF 1]
20	Liqd	8015.33	1250	72.5	...	Liquid
20	Nozl	9.92891	400	1055.4	...	K2A_2"
20	Nozl	27.5826	1000	1114.14	...	N3_6"
20	Nozl	20.7752	1800	1087.15	...	N4_4"
20	Nozl	8.80175	400	-1055.4	...	K2B_2"
20	Wght	40	1000	500	866.025	Lifting Wt 1
20	Wght	40	1000	500	-866.025	Lifting wt 2
30	Liqd	3462.8	540	72.5	...	Liquid 30
30	Nozl	9.92891	300	1055.4	...	K3_2"
30	Nozl	9.92891	600	1055.4	...	K4_2"
30	Wght	80	150	...	...	Clip WT
40	Sadl	319.296	1800	1120	...	New Saddle
40	Liqd	8015.33	1250	72.5	...	Liquid 40
40	Nozl	9.92891	1700	1055.4	...	K1A_2"
40	Nozl	8.34093	2200	-1055.4	...	K1B_2"
40	Nozl	67.7051	700	-1166.53	...	N1_10"
40	Nozl	20.9556	100	-1087.15	...	N2_4"
40	Nozl	19.738	700	1087.15	...	N5_4"
40	Nozl	21.1616	1300	1087.15	...	N6_4"
40	Wght	...	2000	1300	...	PSV
40	Wght	40	1000	500	866.025	Lifting wt 3
40	Wght	40	1000	500	-866.025	Lifting wt 4
50	Liqd	854.572	165.7	72.5	...	Liquid 50

Total Weight of Each Detail Type:

Saddles	638.6
Platforms	2469.1
Liquid	21202.7
Nozzles	281.8
Weights	241.0

Sum of the Detail Weights 24833.3 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	4142.3	4142.3	4142.3	4142.3	4142.3	4142.3
Saddles	638.6	638.6	638.6	638.6	638.6	638.6
Nozzles	281.8	281.8	281.8	281.8	281.8	281.8
Wld Weights	240.5	240.5	240.5	240.5	240.5	240.5



Platforms	...	...	2469.1	2469.1	2469.1	2469.1
Ope Weights	...	...	...	...	...	0.5
Ope. Liquid	...	...	...	...	...	21202.7
Test Liquid	...	22001.0	...	...	...	...
Totals	5303.3	27304.3	7772.4	7772.4	7772.4	28975.6

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	5303.3 kgm
Shop Test Wt.	- Fabricated weight + water (full)	27304.3 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	7772.4 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	7772.4 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	7772.4 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	28975.6 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	27781.9 kgm

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	4403763	4272414
20	30	16336282	16179203
30	40	7057274	6989416
40	50	16336282	16179203
50	60	4403756	4272406

Total	48537356.0	47892640.0 mm ²
Total	48.5	47.9 m ²

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
M1_8"	1.87	1.96	65	150	GR 1.1	...	...	...	...
K2A_2"	1.87	1.96	65	150	GR 1.1	...	...	...	...
N3_6"	1.87	1.96	65	150	GR 1.1	3.180	0.930	1.868	-0.377
N4_4"	4.89	5.10	65	300	GR 1.1	4.062	3.270	4.887	3.270
K2B_2"	1.87	1.96	65	150	GR 1.1	...	...	...	...
K3_2"	1.87	1.96	65	150	GR 1.1	...	...	...	...
K4_2"	1.87	1.96	65	150	GR 1.1	...	...	...	...
K1A_2"	1.87	1.96	65	150	GR 1.1	...	...	...	...
K1B_2"	1.87	1.96	65	150	GR 1.1	...	...	...	...
N1_10"	1.87	1.96	65	150	GR 1.1	2.396	1.713	1.868	0.406
N2_4"	4.89	5.10	65	300	GR 1.1	...	...	...	...
N5_4"	4.89	5.10	65	300	GR 1.1	4.062	3.270	4.887	3.270
N6_4"	4.89	5.10	65	300	GR 1.1	...	...	...	...
Min Rating	0.930	1.960, [for Core Elements]					...	...	-0.377

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N3_6"	196.088	1.20	4512497	4001	0.430	0.930	Pass
N4_4"	138.176	0.50	2011002	2706	0.430	3.270	Pass
N1_10"	302.387	1.20	12507481	6707	0.430	1.713	Pass
N5_4"	138.176	0.50	2011002	2706	0.430	3.270	Pass

Evaluating Flange on Nozzle [N36"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(4512497) + 4 \cdot \text{abs}(4001) \cdot 196.088 \leq \pi \cdot 196.088^3 \cdot ((1.868 - 0.430) + 1.200 \cdot 1.868)$$

$$= 0.75339E+08 \leq 0.87202E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N36" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1.868, 1.200 \cdot 1.868 + 1.868 - 75339304.0 / (\pi \cdot 196.088^3))$$

$$= \min(1.868, 0.930)$$

$$= 0.930 \text{ MPa}$$

Evaluating Flange on Nozzle [N44"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(2011002) + 4 \cdot \text{abs}(2706) \cdot 138.176 \leq \pi \cdot 138.176^3 \cdot ((4.887 - 0.430) + 0.500 \cdot 4.887)$$

$$= 0.33673E+08 \leq 0.57217E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N44" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.887, 0.500 \cdot 4.887 + 4.887 - 33672524.0 / (\pi \cdot 138.176^3))$$

$$= \min(4.887, 3.270)$$

$$= 3.270 \text{ MPa}$$

Evaluating Flange on Nozzle [N110"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(12507481) + 4 \cdot \text{abs}(6707) \cdot 302.387 \leq \pi \cdot 302.387^3 \cdot ((1.868 - 0.430) + 1.200 \cdot 1.868)$$

= 0.20824E+09 ≤ 0.31979E+09 Check Passes

Maximum Pressure for Flange on Nozzle N110" [Pmax]:

= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
 = min(1.868, 1.200 • 1.868 + 1.868 - 208235792.0/(π • 302.387³))
 = min(1.868, 1.713)
 = 1.713 MPa

Evaluating Flange on Nozzle [N54"]:

= 16•ME + 4•FE•G ≤ π•G³[(PR - PD) + FM • PR]
 = 16(2011002)+4•abs(2706)138.176 ≤ π•138.176³[(4.887-0.430)+.500•4.887]
 = 0.33673E+08 ≤ 0.57217E+08 Check Passes

Maximum Pressure for Flange on Nozzle N54" [Pmax]:

= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
 = min(4.887, 0.500 • 4.887 + 4.887 - 33672524.0/(π • 138.176³))
 = min(4.887, 3.270)
 = 3.270 MPa

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class	Class A	
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 3.143

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 0.95 • 1.0 • 0.6 • 57.5²)

= 1876.8 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.7 • 0.738 • 1876.844

= 981.3 N

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	1530.00	1.0000	1.0000	1.0000	57.50	0.7381
20	1530.00	1.0000	1.0000	1.0000	57.50	0.7381
30	1530.00	1.0000	1.0000	1.0000	57.50	0.7381
40	1530.00	1.0000	1.0000	1.0000	57.50	0.7381
50	1530.00	1.0000	1.0000	1.0000	57.50	0.7381

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLAT:[1 OF 1]	9123860.00	1530.00	1.88	17130.24	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Handle Wt	0.00	1530.00	1.88	0.00
Manway davit	0.00	1530.00	1.88	0.00
Lifting Wt 1	0.00	2030.00	1.88	0.00
Lifting wt 2	0.00	2030.00	1.88	0.00
Clip WT	0.00	1530.00	1.88	0.00
PSV	0.00	2830.00	1.88	0.00



Lifting wt 3 0.00 2030.00 1.88 0.00
 Lifting wt 4 0.00 2030.00 1.88 0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1530	2496	708395	1.87756	1079.46
20	30	1530	2496	6240000	1.87756	28351.9
30	40	1530	2496	2695680	1.87756	4107.72
40	50	1530	2496	6240000	1.87756	9508.62
50	60	1530	2496	708169	1.87756	1079.12

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
 Zone Number 2
 Soil Type Rocky or Hard Soil
 Response reduction factor (Force factor) R 2.0000
 Time Period of Vibration (program or user) 0.0303 Seconds
 Value of Sa/g from user Spectrum Data 0.0939
 Percent Damping 2.0 %
 Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.0939 \cdot 1.4) \cdot 1.0$
 $= 0.131$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
 Computed Sa/g = 0.0939

Note:

It was indicated to apply the damping factor, this assumes that the Site Spectrum specified is for 5% damping. If your Sa/g values are already per the actual damping value of the vessel then do not apply the damping factor.

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 38023.84 \cdot 0.1314 \cdot 1.0$
 $= 4996.3 \text{ N}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.1314 \cdot 266167 \cdot 1.0$
 $= 23316.0 \text{ N}$

Earthquake Load Calculation:

	Earthquake	Earthquake	Element
--	------------	------------	---------



From	To	Height mm	Weight N	Ope Load N
10	20	1030	38023.8	4996.28
20	30	1030	38023.8	4996.28
20	30	1030	38023.8	4996.28
30	40	1030	38023.8	4996.28
40	0	1030	38023.8	4996.28
40	50	1030	38023.8	4996.28
50	60	1030	38023.8	4996.28

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3090.000 mm
Center of Gravity of the Platforms	3140.000 mm
Center of Gravity of the Liquid	3089.980 mm
Center of Gravity of the Nozzles	2773.753 mm
Center of Gravity of the Added Weights (Operating)	2780.302 mm
Center of Gravity of the Added Weights (Empty)	2786.804 mm
Center of Gravity of Bare Shell New and Cold	3089.994 mm
Center of Gravity of Bare Shell Corroded	3089.999 mm
Vessel CG in the Operating Condition	3088.533 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	3085.032 mm
Vessel CG in the Test Condition	3088.689 mm

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{ 1036.5 \cdot 7.0 }, 2 \cdot 750.0)$$

$$= 382.8796 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1036.50	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	6180.00	mm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	148.00	degrees
Inside or Mean Depth of Head	Hi or hm	353.74	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1230.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	185157.77	N
Pressure used in Analysis	P	0.389 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	22.84	138.00
Long. Stress at Bottom of Midspan	34.70	138.00
Long. Stress at Top of Saddles	32.10	138.00
Long. Stress at Bottom of Saddles	26.87	138.00
Tangential Shear in Shell	14.68	110.40
Circ. Stress at Horn of Saddle	163.54	172.50
Circ. Compressive Stress in Shell	4.68	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(44126.8 / 2 + 0) \cdot 1230.0 / 2018.6494$$

$$= 40330.9 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(15240.56, 0) \cdot 1230.0 / 4680.0$$

$$= 4005.5 \text{ N}$$

Saddle Reaction Force due to Earthquake FI [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(34973.93, 0) \cdot 1230.0 / 4680.0$$

$$= 9191.9 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(34974/2 + 0) \cdot 1230.0/2018.6494$
 $= 31965.4\ N$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 144827 + \max(4006, 40331, 9192, 31965)$
 $= 185157.8\ N$

Longitudinal Wind Force [FI]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1877.604(2.486 + \pi/4(2.08 \cdot 1.2)^2)$
 $= 15241.352\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	188288.77	N
Transverse Shear Load	22063.40	N
Longitudinal Shear Load	34973.93	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2060.0/2 + 10.0 + 10.0) \sin(148.0/2), 2 \cdot 1036.5)$
 $= 2018.649\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1566	K2 = 0.8177	K3 = 0.5050	K4 = 0.3013
K5 = 0.6777	K6 = 0.0329	K7 = 0.0192	K8 = 0.3052
K9 = 0.2218	K10 = 0.0370	K1* = 0.2730	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -185158 \cdot 750.0 [1 - (1 - 750.0/6180.0 + (1036.5^2 - 353.738^2)/(2 \cdot 750.0 \cdot 6180.0)) / (1 + (4 \cdot 353.74)/(3 \cdot 6180.0))]$
 $= -12298567.0\ N \cdot mm$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L]$
 $= 185158 \cdot 6180/4 [(1 + 2(1037^2 - 354^2)/(6180^2)) / (1 + (4 \cdot 354)/(3 \cdot 6180)) - 4 \cdot 750/6180]$
 $= 140183968.0\ N \cdot mm$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$
 $= 0.389 \cdot 1036.5/(2 \cdot 7.0) - 0.14018E+09/(\pi \cdot 1036.5^2 \cdot 7.0)$
 $= 22.84\ MPa$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 $= 0.389 \cdot 1036.5/(2 \cdot 7.0) + 0.14018E+09/(\pi \cdot 1036.5^2 \cdot 7.0)$
 $= 34.70\ MPa$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$
 $= 0.389 \cdot 1036.5/(2 \cdot 7.0) - 12298567.0/(0.1566 \cdot \pi \cdot 1036.5^2 \cdot 7.0)$
 $= 32.10\ MPa$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 $= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$
 $= 0.389 \cdot 1036.5/(2 \cdot 7.0) + 12298567.0/(0.273 \cdot \pi \cdot 1036.5^2 \cdot 7.0)$
 $= 26.87\ MPa$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L - 2a)/(L + (4 \cdot hm/3))$
 $= 185158(6180.0 - 2 \cdot 750.0) / (6180.0 + (4 \cdot 353.74/3))$

= 130274.2 N

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.8177 \cdot 130274.16 / (1036.5 \cdot 7.0)$$

$$= 14.68 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1036.5 \cdot 7.0)}$$

$$= 66.440 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.6777 \cdot 185158 \cdot 0.1 / (7.0(250.0 + 66.44 + 66.44))$$

$$= -4.68 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -185158 / (4 \cdot 7.0(250.0 + 66.44 + 66.44)) -$$

$$12 \cdot 0.019 \cdot 185158 \cdot 1036.5 / (6180.0 \cdot 7.0^2)$$

$$= -163.54 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(1036.5 \cdot 7.0)}, 2 \cdot 750.0)$$

$$= 382.88 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 4680.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 4680.0(65.0 - 21.1)$$

$$= 2.439 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2030.0001	mm
Baseplate Thickness	Bpthk	19.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	MPa
Height of Web at Center	Hw,c	161.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1033.0 \cdot 7.0)} + 350.0)^3) / 12$$

$$= 13795.874 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 13795.874 + 3378.585(90.182 - 3.5)^2$$

$$= 25399498.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	482.7	7.0	3.5	3378.6	11825.0	0.254E+08
Wearplate	350.0	10.0	12.0	3500.0	42000.0	0.214E+08
Web	10.0	161.0	97.5	1610.0	156975.1	0.356E+07
BasePlate	300.0	19.0	187.5	5700.0	1068750.4	0.542E+08
Totals	...	...	...	14188.6	1279550.5	0.105E+09



Distance to Centroid [C1]:

$$= AY / A$$

$$= 1279551 / 14188.585$$

$$= 90.182 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 148.0 / 2$$

$$= 106.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(106.0) - 0.5 \cdot \sin(106.0)^2) / (\pi - 1.85 + \sin(106.0) \cos(106.0))$$

$$= 0.2556$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.256 \cdot 185157.766$$

$$= 47318.3281 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.3776 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 147.5999 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1230.0 - 1033.0 \cdot \sin(148.0/2) / 1.2915$$

$$= 461.165 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 21830428.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 18.8258 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 163.9998 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(185158 + 3131) \cdot 300.0) / (4 \cdot 2030.0 \cdot 164.0)}$$

$$= 11.281 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 148.0/2) (1030.0 + 10.0 + 10.0)$$

$$= 2018.649 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2018.6494 / (5 - 1)$$

$$= 504.662 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 504.6624 \cdot 300.0 / 2$$

$$= 75699.352 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 188288.766 / (2030.0 \cdot 300.0)$$

$$= 0.309 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 75699.4 \cdot 0.31$$

$$= 23404.496 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2400.0 + 2523.312$$

$$= 4923.312 \text{ mm}^2$$

Compressive Stress [Sc]:

= P/Ar
 = 23404.5/4923.3115
 = 4.754 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	2500.0	...	0.130E+08

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 250.0 - 10.0
 = 240.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/4923.312
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 250.0/2
 = 125.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{13020820/4923.312}$
 = 51.427 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1230.0 - cos(148.0/2)(1030.0 + 10.0 + 10.0) - 19.0
 = 921.581 mm

Intermediate Term [Cc]:

= $\sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$
 = $\sqrt{\frac{2 \cdot \pi^2 \cdot 199955}{246.0}}$
 = 126.667

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 921.581/51.427
 = 17.920

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 34973.9 / (2 • 2030.0) • 504.662 • 921.58/2
 = 2004004.875 N-mm

Compressive Allowable, KL/r < Cc (17.9202 < 126.6671) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (17.92)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (17.92) / (8 \cdot 126.67) - (17.92^3) / (8 \cdot 126.67^3))$
 = 141.6 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 4.75/141.64 + (2004005 • 125.0/13020833) / 164.0
 = 0.151

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	2400.0	0.0	0.130E+08



Web	504.7	10.0	0.0	5046.6	0.0	0.421E+05
Totals	...	...	...	7446.6	...	0.131E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/7446.624$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1230.0 - \cos(1050.0/2)(19.0 + 0.0 + 0.0) - 0.0$$

$$= 161.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(13062041/7446.624)}$$

$$= 41.882 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 161.0/41.882$$

$$= 3.844$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 34973.934 / (2 \cdot 2030.0)$$

$$= 8.614 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 8.614 \cdot 504.662 \cdot 161.0$$

$$= 700198.812 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 700199/104496.43$$

$$= 6.699 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (3.8441 < 126.6671) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.84)^2 / (2 \cdot 126.67^2)) 246 /$$

$$(5/3 + 3 \cdot (3.84) / (8 \cdot 126.67) - (3.84^3) / (8 \cdot 126.67^3))$$

$$= 146.5 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 6.09/146.53 + (700199 \cdot 125.0/13062041) / 164.0$$

$$= 0.082$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.30 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	338.0010 mm ²
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BlArea	247.3890 mm ²
Saddle Load Q0 (Weight)	Q0	147957.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	9191.9 N
Maximum Transverse Force	Ft	22063.4 N
Maximum Longitudinal Force	Fl	34973.9 N

Saddle Bolted to Steel Foundation

No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 $= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 34973.9 / (247.389 \cdot 4)$
 $= 35.3 \text{ MPa. Must be less than } 107.9 \text{ MPa.}$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 22063.4 / (247.389 \cdot 4)$
 $= 22.3 \text{ MPa. Must be less than } 107.9 \text{ MPa.}$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$

Bolt Area due to Shear Load [Bltarears]:
 $= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 34973.93 / (107.87 \cdot 4.0)$
 $= 81.0604 \text{ mm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 1230.0 \cdot 22063.4 + 0.0$
 $= 27148986.00 \text{ N-mm}$

Eccentricity (e):
 $= Rmom / QO$
 $= 27148986 / 147957.88$
 $= 183.42 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[0.0, 81.0604, 0.0]$
 $= 81.0604 \text{ mm}^2$

MDMT Calculations for: Fixed Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 10.0$, $t_r = 3.957$, $c = 3.0 \text{ mm}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.48$, Temp. Reduction = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(250.0 + 1.56 \cdot \sqrt{1036.5 \cdot 7.0}, 2 \cdot 750.0)$
 $= 382.8796 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1036.50	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	6180.00	mm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	250.00	mm



Saddle Bearing Angle delta or theta 148.00 degrees
 Inside or Mean Depth of Head Hi or hm 353.60 mm
 Shell Allowable Stress used in Calculation 138.00 MPa
 Head Allowable Stress used in Calculation 138.00 MPa
 Circumferential Efficiency in Plane of Saddle 1.00
 Circumferential Efficiency at Mid-Span 1.00
 Distance from Saddle Base to Centerline B 1230.00 mm
 Coefficient of Friction mu 0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 184986.98 N
 Pressure used in Analysis P 0.389 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	22.85	138.00
Long. Stress at Bottom of Midspan	34.70	138.00
Long. Stress at Top of Saddles	32.09	138.00
Long. Stress at Bottom of Saddles	26.87	138.00
Tangential Shear in Shell	14.67	110.40
Circ. Stress at Horn of Saddle	163.39	172.50
Circ. Compressive Stress in Shell	4.68	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (44126.8 / 2 + 0) \cdot 1230.0 / 2018.6494$
 $= 40330.9 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(15240.56, 0) \cdot 1230.0 / 4680.0$
 $= 4005.5 \text{ N}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(34973.93, 0) \cdot 1230.0 / 4680.0$
 $= 9191.9 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (34974 / 2 + 0) \cdot 1230.0 / 2018.6494$
 $= 31965.4 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 144656 + \max(4006, 40331, 9192, 31965)$
 $= 184987.0 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi / 4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1877.604 (2.486 + \pi / 4 (2.08 \cdot 1.2)^2)$
 $= 15241.352 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 188118.00 N
 Transverse Shear Load 22063.40 N
 Longitudinal Shear Load 34973.93 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:



$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2060.0/2 + 10.0 + 10.0)\sin(148.0/2), 2 \cdot 1036.5)$$

$$= 2018.649 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1566	K2 = 0.8177	K3 = 0.5050	K4 = 0.3013
K5 = 0.6777	K6 = 0.0329	K7 = 0.0192	K8 = 0.3052
K9 = 0.2218	K10 = 0.0370	K1* = 0.2730	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -184987 \cdot 750.0 [1 - (1 - 750.0/6180.0 + (1036.5^2 - 353.6^2)/(2 \cdot 750.0 \cdot 6180.0))/(1 + (4 \cdot 353.6)/(3 \cdot 6180.0))]$$

$$= -12282359.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 184987 \cdot 6180/4 [(1 + 2(1037^2 - 354^2)/(6180^2))/(1 + (4 \cdot 354)/(3 \cdot 6180)) - 4 \cdot 750/6180]$$

$$= 140063776.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 0.389 \cdot 1036.5/(2 \cdot 7.0) - 0.14006E+09/(\pi \cdot 1036.5^2 \cdot 7.0)$$

$$= 22.85 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.389 \cdot 1036.5/(2 \cdot 7.0) + 0.14006E+09/(\pi \cdot 1036.5^2 \cdot 7.0)$$

$$= 34.70 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.389 \cdot 1036.5/(2 \cdot 7.0) - 12282359.0/(0.1566 \cdot \pi \cdot 1036.5^2 \cdot 7.0)$$

$$= 32.09 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.389 \cdot 1036.5/(2 \cdot 7.0) + 12282359.0/(0.273 \cdot \pi \cdot 1036.5^2 \cdot 7.0)$$

$$= 26.87 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 184987(6180.0 - 2 \cdot 750.0)/(6180.0 + (4 \cdot 353.6/3))$$

$$= 130157.6 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.8177 \cdot 130157.61/(1036.5 \cdot 7.0)$$

$$= 14.67 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1036.5 \cdot 7.0)}$$

$$= 66.440 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.6777 \cdot 184987 \cdot 0.1/(7.0(250.0 + 66.44 + 66.44))$$

$$= -4.68 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ^*7]:

$$= -Q/(4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m/(L \cdot t^2)$$

$$= -184987/(4 \cdot 7.0(250.0 + 66.44 + 66.44)) - 12 \cdot 0.019 \cdot 184987 \cdot 1036.5/(6180.0 \cdot 7.0^2)$$

$$= -163.39 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:



$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(1036.5 \cdot 7.0)}, 2 \cdot 750.0)$$

$$= 382.88 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2030.0001	mm
Baseplate Thickness	Bpthk	19.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	MPa
Height of Web at Center	Hw,c	161.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1033.0 \cdot 7.0)} + 350.0)7.0^3)/12$$

$$= 13795.874 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 13795.874 + 3378.585(90.182 - 3.5)^2$$

$$= 25399498.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	482.7	7.0	3.5	3378.6	11825.0	0.254E+08
Wearplate	350.0	10.0	12.0	3500.0	42000.0	0.214E+08
Web	10.0	161.0	97.5	1610.0	156975.1	0.356E+07
BasePlate	300.0	19.0	187.5	5700.0	1068750.4	0.542E+08
Totals	...	...	...	14188.6	1279550.5	0.105E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1279551/14188.585$$

$$= 90.182 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 148.0/2$$

$$= 106.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(106.0) - 0.5 \cdot \sin(106.0)^2) / (\pi - 1.85 + \sin(106.0) \cos(106.0))$$

$$= 0.2556$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.256 \cdot 184986.984$$

$$= 47274.6875 \text{ N}$$

Tension Stress, St = (Fh/As) = 4.3736 MPa
 Allowed Stress, Sa = 0.6 * Yield Str = 147.5999 MPa

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1230.0 - 1033.0 \cdot \sin(148.0/2) / 1.2915$$

$$= 461.165 \text{ mm}$$

Bending Moment, M = Fh * d = 21810294.0000 N-mm

Bending Stress, Sb = (M * C1 / I) = 18.8084 MPa

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 163.9998 \text{ MPa}$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(184987 + 3131)300.0}{4 \cdot 2030.0 \cdot 164.0}}$$

$$= 11.276 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 148.0/2) (1030.0 + 10.0 + 10.0)$$

$$= 2018.649 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 2018.6494 / (5 - 1)$$

$$= 504.662 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 504.6624 \cdot 300.0 / 2$$

$$= 75699.352 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 188118.0 / (2030.0 \cdot 300.0)$$

$$= 0.309 \text{ N/mm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 75699.4 \cdot 0.31$$

$$= 23383.268 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2400.0 + 2523.312$$

$$= 4923.312 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 23383.3 / 4923.3115$$

$$= 4.750 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	2500.0	...	0.130E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 250.0 - 10.0$$

$$= 240.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 4923.312$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0 / 2$$

$$= 125.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{13020820 / 4923.312}$$

= 51.427 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1230.0 - cos(148.0/2)(1030.0 + 10.0 + 10.0) - 19.0
 = 921.581 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 199955/246.0)}$
 = 126.667

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 921.581/51.427
 = 17.920

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 34973.9 / (2 • 2030.0) • 504.662 • 921.58/2
 = 2004004.875 N-mm

Compressive Allowable, KL/r < Cc (17.9202 < 126.6671) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (17.92)^2 / (2 \cdot 126.67^2)) 246 /$
 $(5/3 + 3 \cdot (17.92) / (8 \cdot 126.67) - (17.92^3) / (8 \cdot 126.67^3))$
 = 141.6 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
 = 4.75/141.64 + (2004005 • 125.0/13020833)/164.0
 = 0.151

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	2400.0	0.0	0.130E+08
Web	504.7	10.0	0.0	5046.6	0.0	0.421E+05
Totals	...	...	...	7446.6	...	0.131E+08

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/7446.624
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 250.0/2
 = 125.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 1230.0 - cos(1050.0/2)(19.0 + 0.0 + 0.0) - 0.0
 = 161.000 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$
 = $\sqrt{(13062041/7446.624)}$
 = 41.882 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 161.0/41.882
 = 3.844

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 34973.934 / (2 • 2030.0)

= 8.614 N/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 8.614 • 504.662 • 161.0
 = 700198.812 N-mm

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 700199/104496.43
 = 6.699 MPa

Compressive Allowable, $KL/r < C_c$ (3.8441 < 126.6671) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)
 = (1-(3.84)²/(2 • 126.67²))246/
 (5/3+3•(3.84)/(8• 126.67)-(3.84³)/(8•126.67³)
 = 146.5 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$
 = 6.08/146.53 + (700199 • 125.0/13062041)/164.0
 = 0.082

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.30 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	338.0010 mm ²
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	247.3890 mm ²
Saddle Load Q0 (Weight)	Q0	147787.1 N
Saddle Load QL (Wind/Seismic contribution)	QL	9191.9 N
Maximum Transverse Force	Ft	22063.4 N
Maximum Longitudinal Force	F1	34973.9 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = 34973.9/(247.389 • 4)
 = 35.3 MPa. Must be less than 107.9 MPa.

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = 22063.4/(247.389 • 4)
 = 22.3 MPa. Must be less than 107.9 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = 34973.93/(107.87 • 4.0)
 = 81.0604 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$
 = 1230.0 • 22063.4 + 0.0
 = 27148986.00 N-mm

Eccentricity (e):

= Rmom / Q0



= 27148986/147787.09
= 183.63 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltarear]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltarear]
= max[0.0, 81.0604, 0.0]
= 81.0604 mm²

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10.0, tr = 3.957, c = 3.0 mm, E* = 0.85
Thickness Ratio = tr * E*/(tg - c) = 0.48, Temp. Reduction = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{ 1035.0 \cdot 10.0 }, 2 \cdot 750.0)$$

$$= 408.7065 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1035.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	6180.00	mm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	148.00	degrees
Inside or Mean Depth of Head	Hi or hm	352.24	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1230.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		156238.94	N
Pressure used in Analysis	P	0.569	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	25.91	234.00
Long. Stress at Bottom of Midspan	32.94	234.00
Long. Stress at Top of Saddles	31.39	234.00
Long. Stress at Bottom of Saddles	28.30	234.00
Tangential Shear in Shell	8.69	138.58
Circ. Stress at Horn of Saddle	70.11	350.99
Circ. Compressive Stress in Shell	2.59	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (14561.8 / 2 + 0) \cdot 1230.0 / 2018.6494$$

$$= 13309.2 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(5029.39, 0) \cdot 1230.0 / 4680.0$$

$$= 1321.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 142930 + \max(1322, 13309, 0, 0)$$

$$= 156238.9 \text{ N}$$

Longitudinal Wind Force [F_L]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1877.604(2.486 + \pi/4(2.08 \cdot 1.2)^2)$
 $= 15241.352 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	159369.94	N
Transverse Shear Load	7280.92	N
Longitudinal Shear Load	5029.39	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2060.0/2 + 10.0 + 10.0)\sin(148.0/2), 2 \cdot 1035.0)$
 $= 2018.649 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1566	K2 = 0.8177	K3 = 0.5050	K4 = 0.3013
K5 = 0.6777	K6 = 0.0329	K7 = 0.0193	K8 = 0.3052
K9 = 0.2218	K10 = 0.0370	K1* = 0.2730	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -156239 \cdot 750.0 [1 - (1 - 750.0/6180.0 + (1035.0^2 - 352.238^2)/(2 \cdot 750.0 \cdot 6180.0))/(1 + (4 \cdot 352.24)/(3 \cdot 6180.0))]$
 $= -10369662.0 \text{ N-mm}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m)/3L) - 4a/L)$
 $= 156239 \cdot 6180.0/4((1 + 2(1035^2 - 352^2)/(6180^2))/(1 + (4 \cdot 352)/(3 \cdot 6180)) - 4 \cdot 750/6180)$
 $= 118336144.0 \text{ N-mm}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$
 $= 0.569 \cdot 1035.0/(2 \cdot 10.0) - 0.11834\text{E}+09/(\pi \cdot 1035.0^2 \cdot 10.0)$
 $= 25.91 \text{ MPa}$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$
 $= 0.569 \cdot 1035.0/(2 \cdot 10.0) + 0.11834\text{E}+09/(\pi \cdot 1035.0^2 \cdot 10.0)$
 $= 32.94 \text{ MPa}$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$
 $= 0.569 \cdot 1035.0/(2 \cdot 10.0) - 10369662.0/(0.1566 \cdot \pi \cdot 1035.0^2 \cdot 10.0)$
 $= 31.39 \text{ MPa}$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 $= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$
 $= 0.569 \cdot 1035.0/(2 \cdot 10.0) + 10369662.0/(0.273 \cdot \pi \cdot 1035.0^2 \cdot 10.0)$
 $= 28.30 \text{ MPa}$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L - 2a)/(L + (4 \cdot h_m/3))$
 $= 156239(6180.0 - 2 \cdot 750.0)/(6180.0 + (4 \cdot 352.24/3))$
 $= 109960.4 \text{ N}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K2 \cdot T / (R_m \cdot t)$
 $= 0.8177 \cdot 109960.38/(1035.0 \cdot 10.0)$
 $= 8.69 \text{ MPa}$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{R_m \cdot t}$
 $= 0.78 \cdot \sqrt{1035.0 \cdot 10.0}$

= 79.353 mm

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.6777 \cdot 156239 \cdot 0.1 / (10.0(250.0 + 79.35 + 79.35))$$

$$= -2.59 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -156239 / (4 \cdot 10.0(250.0 + 79.353 + 79.353)) -$$

$$12 \cdot 0.019 \cdot 156239 \cdot 1035.0 / (6180.0 \cdot 10.0^2)$$

$$= -70.11 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(1035.0 \cdot 10.0)}, 2 \cdot 750.0)$$

$$= 408.71 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2030.0001	mm
Baseplate Thickness	Bpthk	19.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	161.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1030.0 \cdot 10.0)} + 350.0)10.0^3)/12$$

$$= 42360.180 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 42360.18 + 5083.227(83.405 - 5.0)^2$$

$$= 31290522.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	508.3	10.0	5.0	5083.2	25416.1	0.313E+08
Wearplate	350.0	10.0	15.0	3500.0	52500.0	0.164E+08
Web	10.0	161.0	100.5	1610.0	161805.1	0.395E+07
BasePlate	300.0	19.0	190.5	5700.0	1085850.2	0.655E+08
Totals	...	...	...	15893.2	1325571.5	0.117E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1325572 / 15893.227$$

$$= 83.405 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 148.0/2$$

$$= 106.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(106.0) - 0.5 \cdot \sin(106.0)^2) / (\pi - 1.85 + \sin(106.0) \cos(106.0))$$

$$= 0.2556$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.256 \cdot 156238.938$$

$$= 39927.9258 \text{ N}$$



Tension Stress, $St = (Fh/As)$ = 3.6939 MPa
 Allowed Stress, $Sa = 0.6 * Yield Str$ = 155.9972 MPa

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1230.0 - 1030.0 \cdot \sin(148.0/2) / 1.2915$
 $= 463.398 \text{ mm}$

Bending Moment, $M = Fh \cdot d$ = 18510038.0000 N-mm

Bending Stress, $Sb = (M \cdot C1 / I)$ = 13.1692 MPa
 Allowed Stress, $Sa = 2/3 * Yield Str$ = 173.3302 MPa

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + SaddleWt)BasePlateWidth / (4 \cdot BasePlateLength \cdot AllStress)}$
 $= \sqrt{3(156239 + 3131)300.0 / (4 \cdot 2030.0 \cdot 173.33)}$
 $= 10.095 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - Saddle Angle/2) (Inside Radius + Shell Thk + Wear Plate Thk)$
 $= 2 \cdot \cos(90 - 148.0/2) (1030.0 + 10.0 + 10.0)$
 $= 2018.649 \text{ mm}$

Distance between Ribs [e]:
 $= Web Length / (Nribs - 1)$
 $= 2018.6494 / (5 - 1)$
 $= 504.662 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot Bpwid / 2$
 $= 504.6624 \cdot 300.0 / 2$
 $= 75699.352 \text{ mm}^2$

Bearing Pressure [Bp]:
 $= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$
 $= 159369.938 / (2030.0 \cdot 300.0)$
 $= 0.262 \text{ N/mm}^2$

Axial Load [P]:
 $= Ap \cdot Bp$
 $= 75699.4 \cdot 0.26$
 $= 19809.854 \text{ N}$

Area of the Rib and Web [Ar]:
 $= Rib Area + Web Area$
 $= 2400.0 + 2523.312$
 $= 4923.312 \text{ mm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 19809.9 / 4923.3115$
 $= 4.024 \text{ MPa}$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	2500.0	...	0.130E+08

Rib dimension [D]:
 $= Saddle Width - Web Thickness$
 $= 250.0 - 10.0$
 $= 240.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:



$$= AY / A$$

$$= 0.0/4923.312$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(13020820/4923.312)}$$

$$= 51.427 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1230.0 - \cos(148.0/2)(1030.0 + 10.0 + 10.0) - 19.0$$

$$= 921.581 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 921.581/51.427$$

$$= 17.920$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 5029.4 / (2 \cdot 2030.0) \cdot 504.662 \cdot 921.58/2$$

$$= 288183.594 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($17.9202 < 123.2107$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (17.92)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (17.92) / (8 \cdot 123.21) - (17.92^3) / (8 \cdot 123.21^3))$$

$$= 149.5 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.02/149.49 + (288183.59 \cdot 125.0/13020833) / 173.33$$

$$= 0.043$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	2400.0	0.0	0.130E+08
Web	504.7	10.0	0.0	5046.6	0.0	0.421E+05
Totals	...	...	...	7446.6	...	0.131E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/7446.624$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1230.0 - \cos(1050.0/2)(19.0 + 0.0 + 0.0) - 0.0$$

$$= 161.000 \text{ mm}$$

Radius of Gyration [r]:



$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{13062041/7446.624}$$

$$= 41.882 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 161.0/41.882$$

$$= 3.844$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5029.386 / (2 \cdot 2030.0)$$

$$= 1.239 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.239 \cdot 504.662 \cdot 161.0$$

$$= 100691.289 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 100691.289/104496.43$$

$$= 0.963 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($3.8441 < 123.2107$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.84)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (3.84) / (8 \cdot 123.21) - (3.84)^3 / (8 \cdot 123.21^3))$$

$$= 154.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 5.15/154.83 + (100691.29 \cdot 125.0/13062041) / 173.33$$

$$= 0.039$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	338.0010 mm ²
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	247.3890 mm ²
Saddle Load Q0 (Weight)	Q0	146060.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	1321.8 N
Maximum Transverse Force	Ft	7280.9 N
Maximum Longitudinal Force	Fl	5029.4 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5029.4 / (247.389 \cdot 4)$$

$$= 5.1 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 7280.9 / (247.389 \cdot 4)$$

$$= 7.4 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 5029.39 / (103.2 \cdot 4.0)$$

= 12.1848 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1230.0 \cdot 7280.92 + 0.0$$

$$= 8959166.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 8959166/146060.75$$

$$= 61.31 \text{ mm} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 12.1848, 0.0]$$

$$= 12.1848 \text{ mm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{1035.0 \cdot 10.0}, 2 \cdot 750.0)$$

$$= 408.7065 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1035.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	6180.00	mm
Distance from Saddle to Vessel tangent	a1 or a	750.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	148.00	degrees
Inside or Mean Depth of Head	Hi or hm	352.10	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1230.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		156075.45	N
Pressure used in Analysis	P	0.569	MPa

Horizontal Vessel Analysis Results:		Actual MPa	Allowable MPa
Long. Stress at Top of Midspan		25.91	234.00
Long. Stress at Bottom of Midspan		32.94	234.00
Long. Stress at Top of Saddles		31.39	234.00
Long. Stress at Bottom of Saddles		28.30	234.00
Tangential Shear in Shell		8.68	138.58
Circ. Stress at Horn of Saddle		70.04	350.99
Circ. Compressive Stress in Shell		2.59	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(14561.8/2 + 0) \cdot 1230.0/2018.6494$
 $= 13309.2 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(5029.39, 0) \cdot 1230.0/4680.0$
 $= 1321.8 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 142766 + \max(1322, 13309, 0, 0)$
 $= 156075.5 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1877.604(2.486 + \pi/4(2.08 \cdot 1.2)^2)$
 $= 15241.352 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	159206.47	N
Transverse Shear Load	7280.92	N
Longitudinal Shear Load	5029.39	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2060.0/2 + 10.0 + 10.0)\sin(148.0/2), 2 \cdot 1035.0)$
 $= 2018.649 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1566	K2 = 0.8177	K3 = 0.5050	K4 = 0.3013
K5 = 0.6777	K6 = 0.0329	K7 = 0.0193	K8 = 0.3052
K9 = 0.2218	K10 = 0.0370	K1* = 0.2730	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a[1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -156075 \cdot 750.0[1 - (1 - 750.0/6180.0 + (1035.0^2 - 352.1^2)/(2 \cdot 750.0 \cdot 6180.0))/(1 + (4 \cdot 352.1)/(3 \cdot 6180.0))]$
 $= -10354706.0 \text{ N-mm}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 156075 \cdot 6180/4((1 + 2(1035^2 - 352^2)/(6180^2))/(1 + (4 \cdot 352)/(3 \cdot 6180)) - 4 \cdot 750/6180)$
 $= 118219984.0 \text{ N-mm}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 0.569 \cdot 1035.0/(2 \cdot 10.0) - 0.11822E+09/(\pi \cdot 1035.0^2 \cdot 10.0)$
 $= 25.91 \text{ MPa}$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 0.569 \cdot 1035.0/(2 \cdot 10.0) + 0.11822E+09/(\pi \cdot 1035.0^2 \cdot 10.0)$
 $= 32.94 \text{ MPa}$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$
 $= 0.569 \cdot 1035.0/(2 \cdot 10.0) - 10354706.0/(0.1566 \cdot \pi \cdot 1035.0^2 \cdot 10.0)$
 $= 31.39 \text{ MPa}$



Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.569 \cdot 1035.0 / (2 \cdot 10.0) + 10354706.0 / (0.273 \cdot \pi \cdot 1035.0^2 \cdot 10.0)$$

$$= 28.30 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 156075(6180.0 - 2 \cdot 750.0) / (6180.0 + (4 \cdot 352.1 / 3))$$

$$= 109848.4 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.8177 \cdot 109848.36 / (1035.0 \cdot 10.0)$$

$$= 8.68 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1035.0 \cdot 10.0)}$$

$$= 79.353 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.6777 \cdot 156075 \cdot 0.1 / (10.0(250.0 + 79.35 + 79.35))$$

$$= -2.59 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -156075 / (4 \cdot 10.0(250.0 + 79.353 + 79.353)) -$$

$$12 \cdot 0.019 \cdot 156075 \cdot 1035.0 / (6180.0 \cdot 10.0^2)$$

$$= -70.04 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(1035.0 \cdot 10.0)}, 2 \cdot 750.0)$$

$$= 408.71 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2030.0001	mm
Baseplate Thickness	Bpthk	19.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	161.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1030.0 \cdot 10.0)} + 350.0)^3) / 12$$

$$= 42360.180 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 42360.18 + 5083.227(83.405 - 5.0)^2$$

$$= 31290522.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	508.3	10.0	5.0	5083.2	25416.1	0.313E+08
Wearplate	350.0	10.0	15.0	3500.0	52500.0	0.164E+08
Web	10.0	161.0	100.5	1610.0	161805.1	0.395E+07
BasePlate	300.0	19.0	190.5	5700.0	1085850.2	0.655E+08
Totals	...	...	...	15893.2	1325571.5	0.117E+09



Distance to Centroid [C1]:

$$= AY / A$$

$$= 1325572/15893.227$$

$$= 83.405 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 148.0/2$$

$$= 106.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(106.0) - 0.5 \cdot \sin(106.0)^2) / (\pi - 1.85 + \sin(106.0) \cos(106.0))$$

$$= 0.2556$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.256 \cdot 156075.453$$

$$= 39886.1445 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.6901 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1230.0 - 1030.0 \cdot \sin(148.0/2) / 1.2915$$

$$= 463.398 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 18490668.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 13.1554 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}) }$$

$$= \sqrt{ 3(156075 + 3131)300.0 / (4 \cdot 2030.0 \cdot 173.33) }$$

$$= 10.090 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 148.0/2) (1030.0 + 10.0 + 10.0)$$

$$= 2018.649 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2018.6494 / (5 - 1)$$

$$= 504.662 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 504.6624 \cdot 300.0/2$$

$$= 75699.352 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 159206.469 / (2030.0 \cdot 300.0)$$

$$= 0.261 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 75699.4 \cdot 0.26$$

$$= 19789.533 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2400.0 + 2523.312$$

$$= 4923.312 \text{ mm}^2$$

Compressive Stress [Sc]:

= P/A_r
 = 19789.5/4923.3115
 = 4.020 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	2500.0	...	0.130E+08

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 250.0 - 10.0
 = 240.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/4923.312
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 250.0/2
 = 125.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{13020820/4923.312}$
 = 51.427 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2) \cdot (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 = 1230.0 - $\cos(148.0/2) \cdot (1030.0 + 10.0 + 10.0) - 19.0$
 = 921.581 mm

Intermediate Term [Cc]:

= $\sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$
 = $\sqrt{\frac{2 \cdot \pi^2 \cdot 199955}{260.0}}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 921.581/51.427
 = 17.920

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 = 5029.4 / (2 • 2030.0) • 504.662 • 921.58/2
 = 288183.594 N-mm

Compressive Allowable, $KL/r < C_c$ (17.9202 < 123.2107) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 = $(1 - (17.92)^2 / (2 \cdot 123.21^2)) 260 /$
 = $(5/3 + 3 \cdot (17.92) / (8 \cdot 123.21) - (17.92^3) / (8 \cdot 123.21^3))$
 = 149.5 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $Sc/Sca + (Rm \cdot C1 / I) / Sba$
 = 4.02/149.49 + (288183.59 • 125.0/13020833)/173.33
 = 0.043

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	2400.0	0.0	0.130E+08



Web	504.7	10.0	0.0	5046.6	0.0	0.421E+05
Totals	...	...	...	7446.6	...	0.131E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/7446.624$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1230.0 - \cos(1050.0/2)(19.0 + 0.0 + 0.0) - 0.0$$

$$= 161.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(13062041/7446.624)}$$

$$= 41.882 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 161.0/41.882$$

$$= 3.844$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5029.386/(2 \cdot 2030.0)$$

$$= 1.239 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.239 \cdot 504.662 \cdot 161.0$$

$$= 100691.289 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 100691.289/104496.43$$

$$= 0.963 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (3.8441 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.84)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (3.84) / (8 \cdot 123.21) - (3.84)^3 / (8 \cdot 123.21^3))$$

$$= 154.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 5.15/154.83 + (100691.29 \cdot 125.0/13062041) / 173.33$$

$$= 0.039$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	338.0010 mm ²
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BlArea	247.3890 mm ²
Saddle Load Q0 (Weight)	Q0	145897.3 N
Saddle Load QL (Wind/Seismic contribution)	QL	1321.8 N
Maximum Transverse Force	Ft	7280.9 N
Maximum Longitudinal Force	Fl	5029.4 N
Saddle Bolted to Steel Foundation	No	



Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $5029.4 / (247.389 \cdot 4)$
= 5.1 MPa. Must be less than 103.2 MPa.

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $7280.9 / (247.389 \cdot 4)$
= 7.4 MPa. Must be less than 103.2 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [$Bltarearl$]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [$Bltarears$]:
= $F_l / (\text{BoltShearAllowable} \cdot Nbolts)$
= $5029.39 / (103.2 \cdot 4.0)$
= 12.1848 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [$Rmom$]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1230.0 \cdot 7280.92 + 0.0$
= 8959166.00 N-mm

Eccentricity (e):
= $Rmom / QO$
= $8959166 / 145897.27$
= 61.38 mm < $B_{plen}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [$Bltareart$]:
= 0 (No Uplift)

Required Area of a Single Bolt [$Bltarear$]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 12.1848, 0.0]$
= 12.1848 mm²

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: M18"

From: 10

Pressure for Reinforcement Calculations	P	0.3886	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Crown Radius of Torispherical Head	L	2077.0000	mm
Inside Knuckle Radius of Torispherical Head	r	124.6200	mm
Head Finished (Minimum) Thickness	t	10.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M18"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	8.625	in.
Actual Thickness Used in Calculation	0.282	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Torispherical Head, tr [Int. Press]
 $= P \cdot L \cdot M / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ Appendix 1-4 (d)
 $= 0.39 \cdot 2080.0 \cdot 1.0 / (2 \cdot 138 \cdot 1.0 - 0.2 \cdot 0.39)$
 $= 2.9291 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]



$$= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 0.39 \cdot 109.5375 / (118 \cdot 1.0 + 0.4 \cdot 0.39)$$

$$= 0.3602 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	415.5242	mm
Parallel to Vessel Wall, opening length	d	207.7621	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	14.1411	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, Sn/Sv)$$

$$= \min(1, 118.0/138.0)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, Sn/Sv)$$

$$= \min(1, 118.0/138.0)$$

$$= 0.855$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(0.855, 1.0)$$

$$= 0.855$$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	613.361	NA	NA
Area in Shell	A1	839.102	NA	NA
Area in Nozzle Wall	A2	128.083	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.509	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	1052.693	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (207.7621 \cdot 2.9291 \cdot 1.0 + 2 \cdot 5.6564 \cdot 2.9291 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 613.361 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 207.762(1.0 \cdot 7.0 - 1.0 \cdot 2.929) - 2 \cdot 5.656$$

$$(1.0 \cdot 7.0 - 1.0 \cdot 2.9291) \cdot (1 - 0.855)$$

$$= 839.102 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 14.14)(5.66 - 0.36)0.8551$$

$$= 128.083 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$$

$$= 85.509 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 1.8602 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 1.8602 \text{ mm}$$

Available Nozzle Neck Thickness = 7.1564 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 7.156$, $t_r = 0.36$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.064$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 7.156$, $t_r = 0.36$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.064$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.39/1.96 = 0.198$

Weld Size Calculations, Description: M18"

Intermediate Calc. for nozzle/shell Welds T_{min} 5.6564 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.9595 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (613.3612 - 839.1016 + 2 \cdot 5.6564 \cdot 0.8551 \cdot (1.0 \cdot 7.0 - 2.9291)) \cdot 138)$
 $= \max(0, -25715.10) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (128.0831 + 0.0 + 85.5088 - 0.0 \cdot 0.86) \cdot 138$
 $= 29472.64 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (128.0831 + 0.0 + 85.5088 + (67.7147)) \cdot 138$
 $= 38816.30 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (128.0831 + 0.0 + 85.5088 + 0.0 + (67.7147)) \cdot 138$
 $= 38816.30 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= $(\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
= $(3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$
= 198955. N

Shear, Nozzle Wall [Snw]:
= $(\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
= $(3.1416 \cdot 106.7093) \cdot (7.1564 - 1.5) \cdot 0.7 \cdot 118$
= 156617. N

Tension, Shell Groove Weld [Tngw]:
= $(\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
= $(3.1416/2.0) \cdot 219.075 \cdot (10.0 - 3.0) \cdot 0.74 \cdot 138$
= 245967. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (198955 + 156617) = 355572 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (198955 + 0 + 245967 + 0) = 444921 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (198955 + 245967 + 0) = 444921 N

Summary of Failure Path Calculations:

Path 1-1 = 355571 N	, must exceed W = 0 N	or W1 = 29472 N
Path 2-2 = 444921 N	, must exceed W = 0 N	or W2 = 38816 N
Path 3-3 = 444921 N	, must exceed W = 0 N	or W3 = 38816 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.8766 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 162.8765 mm

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Input, Nozzle Desc: K2A2"

From: 20

Pressure for Reinforcement Calculations	P	0.3800	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2060.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		450.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	280.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K2A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)



$$= 0.38 \cdot 1033.0 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$$

$$= 2.8494 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.38 \cdot 26.9 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$
 $= 0.0742 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K2A2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.5742	mm
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.8494	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.8494	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(5.8494, 5.5)]$
 $= 5.8494 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.5742, 5.8494)$
 $= 5.8494 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10.0, tr = 2.849, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.407$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.38 / 1.96 = 0.194$

Weld Size Calculations, Description: K2A2"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot t_{min}$	$8.4840 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles



exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7334 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 290.7334 mm

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Input, Nozzle Desc: N36"

From: 20

Pressure for Reinforcement Calculations	P	0.3800	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2060.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1050.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	230.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	328.2750	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		80.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625	in.
Actual Thickness Used in Calculation	0.378	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.38 \cdot 1033.0 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$
 $= 2.8494 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.38 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 0.38)$
 $= 0.2706 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 304.1452 mm
 Parallel to Vessel Wall, opening length d 152.0726 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 17.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	440.007	NA	NA
Area in Shell	A1	621.446	NA	NA
Area in Nozzle Wall	A2	234.354	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		80.164	NA	NA
Area in Element	A5	1358.702	NA	NA
TOTAL AREA AVAILABLE	Atot	2294.667	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (152.0726 \cdot 2.8494 \cdot 1.0 + 2 \cdot 8.1012 \cdot 2.8494 \cdot 1.0 \cdot (1 - 0.86))$
 $= 440.007 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152.073(1.0 \cdot 7.0 - 1.0 \cdot 2.849) - 2 \cdot 8.101$
 $(1.0 \cdot 7.0 - 1.0 \cdot 2.8494) \cdot (1 - 0.855)$



= 621.446 mm²

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 17.5) \cdot (8.1 - 0.27) \cdot 0.8551$$

$$= 234.354 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4$$

$$= (93.75) \cdot 0.86 + (0.0) \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 80.164 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (304.1452 - 168.275) \cdot 10.0 \cdot 1.0$$

$$= 1358.702 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7706 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 5.8494 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.8494 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 7.7200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.72, \max(5.8494, 5.5)]$$

$$= 5.8494 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.7706, 5.8494)$$

$$= 5.8494 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29.7,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 265.3 MPa	Passed
Occasional	: 1.7,	Allowable	: 156.9 MPa	Passed
Shear	: 19.7,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.601, tr = 0.271, c = 1.5 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.033, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.601, tr = 0.271, c = 1.5 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.033, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 0.271, c = 1.5 mm, E* = 1.0



Thickness Ratio = $tr * E^* / (tg - c) = 0.033$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 10.0$, $tr = 2.849$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.407$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 9.601$, $tr = 0.271$, $c = 1.5$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.033$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.38/1.96 = 0.194$

Weld Size Calculations, Description: N36"

Intermediate Calc. for nozzle/shell Welds T_{min} 8.1012 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 7.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6708 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$3.5000 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (440.0071 - 621.4459 + 2 \cdot 8.1012 \cdot 0.8551 \cdot (1.0 \cdot 7.0 - 2.8494)) \cdot 138)$
 $= \max(0, -17101.21) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (234.3543 + 1358.7018 + 80.1645 - 0.0 \cdot 0.86) \cdot 138$
 $= 230880.61 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (234.3543 + 0.0 + 85.5088 + (96.9813)) \cdot 138$
 $= 57518.59 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (234.3543 + 0.0 + 80.1645 + 1358.7018 + (96.9813)) \cdot 138$
 $= 244262.67 \text{ N}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 152820. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 328.275 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 348649. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 80.0869) \cdot (9.6012 - 1.5) \cdot 0.7 \cdot 118$
 $= 168346. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 168.275 \cdot 10.0 \cdot 0.74 \cdot 138$
 $= 269906. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 168.275 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 161941. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (348649 + 168346) = 516995 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (152820 + 269907 + 161941 + 0) = 584668 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (348649 + 161941 + 0) = 510590 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 516995 N , must exceed W = 0 N	or W1 = 230880 N
Path 2-2 = 584667 N , must exceed W = 0 N	or W2 = 57518 N
Path 3-3 = 510589 N , must exceed W = 0 N	or W3 = 244262 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.4422 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4422 mm

Input Echo, WRC297 Item 1, Description: N36" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2060.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm



Nozzle Diameter	168.275	mm
Nozzle Thickness	9.6012	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	328.275	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	0.38	MPa
Radial Load	P	-4000.85	N
Circumferential Shear	Vc	6001.29	N
Longitudinal Shear	Vl	6001.29	N
Circumferential Moment	Mc	0.271E+07	N-mm
Longitudinal Moment	Ml	0.361E+07	N-mm
Torsional Moment	Mt	0.450E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	Yes
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Indices for Nozzles in Cylindrical Shells per Table 5-D.2

Stress	Longitudinal Plane		Transverse Plane	
	Inside Corner	Outside Corner	Inside Corner	Outside Corner
sn	3.100	1.200	1.000	2.100
st	-0.200	1.000	-0.200	2.600

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(1041.5(10.0 - 3.0))}$
 $= 450.044 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2083.000 mm
Nozzle Outside Diameter	(d) =	168.275 mm
Vessel Thickness used	(T) =	17.000 mm
Nozzle Thickness used	(t) =	8.101 mm
T / t	=	2.098
d / t	=	20.772

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.894$$

Nr/P	=	0.109
Mr/P	=	0.136
M0/P	=	0.089
N0/P	=	0.181
MrD/Mc	=	0.215
NrDL/Mc	=	0.149
M0d/Mc	=	0.159
N0DL/Mc	=	0.074
MrD/Ml	=	0.175
NrDL/Ml	=	0.127
M0D/Ml	=	0.123
N0DL/Ml	=	0.189



Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-54	54	54	-54
Normal Pressure Stress	27	72	27	72
Outplane Stress Summary	-31	108	106	26

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-77	77	77	-77
Inplane Pressure Stress	23	-4	23	-4
Inplane Stress Summary	-51	53	123	-82

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	7	7	4	4

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	53	109	124	109

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-53	53	53	-53
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	48	23	48	23
Outplane Stress Summary	1	67	116	-30

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	11	-11	11	-11



Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-71	71	71	-71
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	60	-4	60	-4

Inplane Stress Summary	-6	49	153	-77
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	4	4	7	7
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	12	68	154	79
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-14	-14	14	14
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	4	4	4

Hoop Stress Summary	-7	-7	20	21
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	29	-29	29	-29
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-23	-23	23	23
Axial Bending (ML)	-218	218	218	-218
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-208	168	274	-222
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12

Shear Stress Summary	15	15	9	9
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside



Two * Max Shear Stress 209 177 274 243

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	4	4	4
Hoop Stress Summary	2	2	10	11

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	29	-29	29	-29
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-206	206	206	-206
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-191	162	257	-216

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	12	12	12	12
Shear Stress Summary	9	9	15	15

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	194	163	258	229

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(1036.5(10.0 - 3.0))}$$

$$= 449.366 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2073.000 mm
 Nozzle Outside Diameter (d) = 328.275 mm
 Vessel Thickness used (T) = 7.000 mm
 Nozzle Thickness used (t) = 88.101 mm
 $T / t = 0.079$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.725$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.069



Mr/P = 0.116
 M0/P = 0.034
 N0/P = 0.039
 MrD/Mc = 0.276
 NrDL/Mc = 0.116
 M0d/Mc = 0.083
 N0DL/Mc = 0.058
 MrD/Ml = 0.086
 NrDL/Ml = 0.096
 M0D/Ml = 0.025
 N0DL/Ml = 0.053

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-11	-11	11	11
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	55	56	55	56
Outplane Stress Summary	29	65	121	20

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-21	-21	21	21
Inplane Bending (ML)	-115	115	115	-115
Inplane Pressure Stress	27	27	27	27
Inplane Stress Summary	-46	70	227	-117

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	76	74	227	137

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-83	83	83	-83
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	27	27	27	27

Outplane Stress Summary	-45	88	141	-59
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	-19	-19	19	19
Inplane Bending (Mc)	-278	278	278	-278
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	55	56	55	56
Inplane Stress Summary	-179	263	416	-253

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	5	5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	179	263	416	254

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	27	72	27	72	60	-4	60	-4
Circ. Pl (SUS)	-12	-12	16	16	-7	-7	9	9
Circ. Q (SUS)	-47	47	61	-61	-60	60	82	-82
Long. Pm (SUS)	23	-4	23	-4	48	23	48	23
Long. Pl (SUS)	-8	-8	10	10	-2	-2	6	6
Long. Q (SUS)	-66	66	88	-88	-46	46	60	-60
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	27.0	76.0	27.0	76.0	60.0	27.0	60.0	27.0
Pm+Pl (SUS)	16.0	72.0	43.1	88.0	53.1	32.1	69.1	29.0
Pm+Pl+Q (Total)	52.7	107.7	121.9	109.3	10.6	67.8	151.9	77.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	76.00	138.00	Passed
Pm+Pl (SUS)	88.01	207.00	Passed
Pm+Pl+Q (TOTAL)	151.95	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	55	56	55	56	55	56	55	56
Circ. Pl (SUS)	-8	-8	14	14	-14	-14	24	24
Circ. Q (SUS)	-18	18	50	-50	-222	222	334	-334
Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-16	-16	26	26	-6	-6	12	12
Long. Q (SUS)	-59	59	171	-171	-67	67	99	-99
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	55.0	56.0	55.0	56.0	55.0	56.0	55.0	56.0
Pm+Pl (SUS)	47.0	48.0	69.1	70.1	41.0	42.0	79.0	80.0
Pm+Pl+Q (Total)	77.4	72.5	224.0	138.1	181.0	264.0	413.1	254.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	56.00	138.00	Passed
Pm+Pl (SUS)	80.02	207.00	Passed
Pm+Pl+Q (TOTAL)	413.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	4	4	4	4	4	4	4
Circ. Pl (SUS)	-12	-12	16	16	-2	-2	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-23	-23	23	23	-17	-17	17	17
Long. Q (SUS)	-189	189	247	-247	-177	177	235	-235
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Pm+Pl (SUS)	21.3	21.3	25.7	25.7	17.5	17.5	19.4	19.4
Pm+Pl+Q (Total)	211.0	178.2	272.4	242.8	195.0	162.6	254.8	227.7

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	118.00	Passed
Pm+Pl (SUS)	25.70	177.00	Passed
Pm+Pl+Q (TOTAL)	272.40	354.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N44"

From: 20

Pressure for Reinforcement Calculations	P	0.3800	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2060.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1850.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	230.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	6.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	214.3000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N44"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500	in.
Actual Thickness Used in Calculation	0.295	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.38 \cdot 1033.0 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$
 $= 2.8494 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.38 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.38)$
 $= 0.1838 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 204.6407 mm
 Parallel to Vessel Wall, opening length d 102.3204 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 17.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	296.499	NA	NA
Area in Shell	A1	417.485	NA	NA
Area in Nozzle Wall	A2	173.763	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		30.783	NA	NA
Area in Element	A5	903.407	NA	NA
TOTAL AREA AVAILABLE	Atot	1525.437	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (102.3204 \cdot 2.8494 \cdot 1.0 + 2 \cdot 5.9898 \cdot 2.8494 \cdot 1.0 \cdot (1 - 0.86))$
 $= 296.499 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 102.32(1.0 \cdot 7.0 - 1.0 \cdot 2.849) - 2 \cdot 5.99$
 $(1.0 \cdot 7.0 - 1.0 \cdot 2.8494) \cdot (1 - 0.855)$



= 417.485 mm²

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot f_{r2}$$

$$= (2 \cdot 17.5) \cdot (5.99 - 0.18) \cdot 0.8551$$

$$= 173.763 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot f_{r3} + (W_i - c_{an}/0.707)^2 \cdot f_{r2} + W_p^2 \cdot f_{r4}$$

$$= 6.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 30.783 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (204.6407 - 114.3) \cdot 10.0 \cdot 1.0$$

$$= 903.407 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 1.6838 \text{ mm}$
Wall Thickness per UG16(b),	$t_{r16b} = 5.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 5.8494 \text{ mm}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 5.8494 \text{ mm}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 5.5000 \text{ mm}$
Wall Thickness per table UG-45	$t_{b3} = 6.7578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[6.7578, \max(5.8494, 5.5)]$$

$$= 5.8494 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.6838, 5.8494)$$

$$= 5.8494 \text{ mm}$$

Available Nozzle Neck Thickness = 7.4898 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 38.5,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 256.5 MPa	Passed
Occasional	: 1.5,	Allowable	: 156.9 MPa	Passed
Shear	: 26.2,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 7.49$, $t_r = 0.184$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.031$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 7.49$, $t_r = 0.184$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.031$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 7.49$, $t_r = 0.184$, $c = 1.5 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.031$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 10.0$, $tr = 2.849$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.407$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 7.49$, $tr = 0.184$, $c = 1.5$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.031$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.38/5.10 = 0.075$

Weld Size Calculations, Description: N44"

Intermediate Calc. for nozzle/shell Welds T_{min} 5.9898 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 7.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.1929 = 0.7 \cdot t_{min}$	$4.2420 = 0.7 \cdot W_o$ mm
Pad Weld	$3.5000 = 0.5 \cdot T_{minPad}$	$7.0700 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (296.499 - 417.4845 + 2 \cdot 5.9898 \cdot 0.8551 \cdot (1.0 \cdot 7.0 - 2.8494)) \cdot 138)$
 $= \max(0, -10827.51) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (173.7627 + 903.407 + 30.7832 - 0.0 \cdot 0.86) \cdot 138$
 $= 152881.73 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (173.7627 + 0.0 + 30.7832 + (71.7056)) \cdot 138$
 $= 38118.77 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (173.7627 + 0.0 + 30.7832 + 903.407 + (71.7056)) \cdot 138$
 $= 162776.06 \text{ N}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 114.3 \cdot 6.0 \cdot 0.49 \cdot 118$
 $= 62281. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 214.3 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 227600. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 54.1551) \cdot (7.4898 - 1.5) \cdot 0.7 \cdot 118$
 $= 84168. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 114.3 \cdot 10.0 \cdot 0.74 \cdot 138$
 $= 183333. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 114.3 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 109998. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (227600 + 84168) = 311768 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (62281 + 183333 + 109998 + 0) = 355612 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (227600 + 109998 + 0) = 337598 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 311767 N , must exceed W = 0 N	or W1 = 152881 N
Path 2-2 = 355611 N , must exceed W = 0 N	or W2 = 38118 N
Path 3-3 = 337597 N , must exceed W = 0 N	or W3 = 162776 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5867 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.5867 mm

Input Echo, WRC297 Item 1, Description: N44" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2060.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm



Nozzle Diameter	114.300	mm
Nozzle Thickness	7.4898	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	214.300	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	0.38	MPa
Radial Load	P	-2706.46	N
Circumferential Shear	Vc	4000.85	N
Longitudinal Shear	Vl	4000.85	N
Circumferential Moment	Mc	0.121E+07	N-mm
Longitudinal Moment	Ml	0.161E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	Yes
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Indices for Nozzles in Cylindrical Shells per Table 5-D.2

Stress	Longitudinal Plane		Transverse Plane	
	Inside Corner	Outside Corner	Inside Corner	Outside Corner
sn	3.100	1.200	1.000	2.100
st	-0.200	1.000	-0.200	2.600

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1041.5(10.0 - 3.0))}$
 $= 396.069 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2083.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	17.000	mm
Nozzle Thickness used	(t) =	5.990	mm
T / t	=	2.838	
d / t	=	19.082	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.607$$

Nr/P	=	0.130
Mr/P	=	0.146
M0/P	=	0.129
N0/P	=	0.240
MrD/Mc	=	0.192
NrDL/Mc	=	0.150
M0d/Mc	=	0.203
N0DL/Mc	=	0.124
MrD/Ml	=	0.177
NrDL/Ml	=	0.142
M0D/Ml	=	0.181
N0DL/Ml	=	0.141



Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-52	52	52	-52
Normal Pressure Stress	27	72	27	72
Outplane Stress Summary	-22	113	97	21

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-51	51	51	-51
Inplane Pressure Stress	23	-4	23	-4
Inplane Stress Summary	-25	33	91	-56

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	7	7	4	4

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	31	113	99	77

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	48	23	48	23
Outplane Stress Summary	9	58	107	-21

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8



Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-42	42	42	-42
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	60	-4	60	-4

Inplane Stress Summary	22	25	117	-48
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	4	4	7	7
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	23	58	121	50
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4

Hoop Stress Summary	0	0	12	13
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	36	-36	36	-36
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-30	-30	30	30
Axial Bending (ML)	-249	249	249	-249
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-240	185	319	-251
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	16	16	16	16

Shear Stress Summary	19	19	12	12
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside



Two * Max Shear Stress 242 189 319 266

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4
Hoop Stress Summary	1	1	10	10

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	36	-36	36	-36
Axial Membrane (Mc)	-22	-22	22	22
Axial Bending (Mc)	-206	206	206	-206
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-190	150	269	-217

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	16	16	16	16
Shear Stress Summary	12	12	19	19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	193	151	270	231

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1036.5(10.0 - 3.0))}$
 $= 395.391 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2073.000 mm
Nozzle Outside Diameter	(d) =	214.300 mm
Vessel Thickness used	(T) =	7.000 mm
Nozzle Thickness used	(t) =	55.990 mm
T / t	=	0.125
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.779$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

$$Nr/P = 0.089$$



Mr/P = 0.159
 M0/P = 0.047
 N0/P = 0.064
 MrD/Mc = 0.295
 NrDL/Mc = 0.105
 M0d/Mc = 0.090
 N0DL/Mc = 0.061
 MrD/Ml = 0.122
 NrDL/Ml = 0.112
 M0D/Ml = 0.037
 N0DL/Ml = 0.083

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	55	56	55	56
Outplane Stress Summary	27	66	121	22

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-17	-17	17	17
Inplane Bending (ML)	-112	112	112	-112
Inplane Pressure Stress	27	27	27	27
Inplane Stress Summary	-43	75	214	-115

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	72	77	215	138

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-62	62	62	-62
Outplane Membrane (ML)	0	0	0	0

Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	27	27	27	27

Outplane Stress Summary	-22	71	116	-39
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-203	203	203	-203
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	55	56	55	56
Inplane Stress Summary	-101	199	328	-182

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	5	5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	101	199	328	182

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	27	72	27	72	60	-4	60	-4
Circ. Pl (SUS)	-4	-4	8	8	-4	-4	6	6
Circ. Q (SUS)	-45	45	59	-59	-34	34	50	-50
Long. Pm (SUS)	23	-4	23	-4	48	23	48	23
Long. Pl (SUS)	-5	-5	7	7	-2	-2	6	6
Long. Q (SUS)	-43	43	59	-59	-37	37	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	27.0	76.0	27.0	76.0	60.0	27.0	60.0	27.0
Pm+Pl (SUS)	23.2	77.0	35.2	80.0	56.1	29.1	66.1	29.0
Pm+Pl+Q (Total)	29.7	113.5	96.2	77.4	23.1	58.5	118.6	49.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	76.00	138.00	Passed
Pm+Pl (SUS)	80.01	207.00	Passed
Pm+Pl+Q (TOTAL)	118.64	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	55	56	55	56	55	56	55	56
Circ. Pl (SUS)	-9	-9	15	15	-8	-8	16	16
Circ. Q (SUS)	-19	19	49	-49	-151	151	255	-255
Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-13	-13	21	21	-4	-4	10	10
Long. Q (SUS)	-60	60	164	-164	-47	47	77	-77
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	55.0	56.0	55.0	56.0	55.0	56.0	55.0	56.0
Pm+Pl (SUS)	46.0	47.0	70.0	71.0	47.0	48.0	71.0	72.0
Pm+Pl+Q (Total)	73.4	75.7	212.0	138.1	104.0	199.0	326.1	183.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	56.00	138.00	Passed
Pm+Pl (SUS)	72.03	207.00	Passed
Pm+Pl+Q (TOTAL)	326.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	4	3	4	3	4	3	4
Circ. Pl (SUS)	-4	-4	8	8	-2	-2	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-29	-29	31	31	-21	-21	23	23
Long. Q (SUS)	-213	213	285	-285	-170	170	242	-242
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	16	16	16	16	16	16	16	16
Pm (SUS)	3.0	4.0	3.0	4.0	3.0	4.0	3.0	4.0
Pm+Pl (SUS)	28.3	28.6	32.4	32.4	21.8	22.8	24.6	24.6
Pm+Pl+Q (Total)	243.0	188.9	317.6	266.3	192.8	151.1	267.4	231.1



Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	118.00	Passed
Pm+Pl (SUS)	32.44	177.00	Passed
Pm+Pl+Q (TOTAL)	317.55	354.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K2B2"

From: 20

Pressure for Reinforcement Calculations	P	0.3987	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2060.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		450.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	230.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K2B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)



$$= 0.4 \cdot 1033.0 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$$

$$= 2.9894 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.4 \cdot 26.9 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
 $= 0.0778 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2B2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.5778	mm
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.9894	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.9894	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(5.9894, 5.5)]$
 $= 5.9894 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.5778, 5.9894)$
 $= 5.9894 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10.0, tr = 2.989, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.427$, Temp. Reduction = 44 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.40 / 1.96 = 0.203$

Weld Size Calculations, Description: K2B2"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot t_{min}$	$8.4840 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles



exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7334 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 240.7334 mm

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Input, Nozzle Desc: K32"

From: 30

Pressure for Reinforcement Calculations	P	0.3800	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2060.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2850.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	280.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)



$$= 0.38 \cdot 1033.0 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$$

$$= 2.8494 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.38 \cdot 26.9 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$
 $= 0.0742 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.5742	mm
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.8494	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.8494	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(5.8494, 5.5)]$
 $= 5.8494 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.5742, 5.8494)$
 $= 5.8494 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10.0, tr = 2.849, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.407$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.38 / 1.96 = 0.194$

Weld Size Calculations, Description: K32"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot t_{min}$	$8.4840 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles



exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7334 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 290.7334 mm

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Input, Nozzle Desc: K42"

From: 30

Pressure for Reinforcement Calculations	P	0.3800	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2060.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3150.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	280.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)



$$= 0.38 \cdot 1033.0 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$$

$$= 2.8494 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.38 \cdot 26.9 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$
 $= 0.0742 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K42".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.5742	mm
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.8494	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.8494	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(5.8494, 5.5)]$
 $= 5.8494 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.5742, 5.8494)$
 $= 5.8494 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10.0, tr = 2.849, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.407$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.38 / 1.96 = 0.194$

Weld Size Calculations, Description: K42"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm
---	------	--------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot t_{min}$	$8.4840 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles



exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7334 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 290.7334 mm

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Input, Nozzle Desc: K1A2"

From: 40

Pressure for Reinforcement Calculations	P	0.3800	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2060.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5330.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	280.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K1A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)



$$= 0.38 \cdot 1033.0 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$$

$$= 2.8494 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.38 \cdot 26.9 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$
 $= 0.0742 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.5742	mm
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.8494	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.8494	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(5.8494, 5.5)]$
 $= 5.8494 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.5742, 5.8494)$
 $= 5.8494 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10.0, tr = 2.849, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.407$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.38 / 1.96 = 0.194$

Weld Size Calculations, Description: K1A2"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot t_{min}$	$8.4840 = 0.7 \cdot W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles



exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7334 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 290.7334 mm

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Input, Nozzle Desc: K1B2"

From: 40

Pressure for Reinforcement Calculations	P	0.3987	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2060.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5830.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K1B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)



$$= 0.4 \cdot 1033.0 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$$

$$= 2.9894 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.4 \cdot 26.9 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
 $= 0.0778 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1B2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.5778	mm
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 5.9894	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.9894	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(5.9894, 5.5)]$
 $= 5.9894 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.5778, 5.9894)$
 $= 5.9894 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 10.0, tr = 2.989, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.427$, Temp. Reduction = 44 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.40 / 1.96 = 0.203$

Weld Size Calculations, Description: K1B2"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot t_{min}$	$8.4840 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles



exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7334 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.2834 mm

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Input, Nozzle Desc: N110"

From: 40

Pressure for Reinforcement Calculations	P	0.3987	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2060.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4330.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	573.0500	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		150.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N110"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750	in.
Actual Thickness Used in Calculation	0.520	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.4 \cdot 1033.0 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
 $= 2.9894 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.4 \cdot 136.525 / (118 \cdot 1.0 + 0.4 \cdot 0.4)$
 $= 0.4606 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 499.2934 mm
 Parallel to Vessel Wall, opening length d 249.6467 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 17.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	756.444	NA	NA
Area in Shell	A1	987.620	NA	NA
Area in Nozzle Wall	A2	336.422	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		80.164	NA	NA
Area in Element	A5	2262.434	NA	NA
TOTAL AREA AVAILABLE	Atot	3666.641	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (249.6467 \cdot 2.9894 \cdot 1.0 + 2 \cdot 11.7016 \cdot 2.9894 \cdot 1.0 \cdot (1 - 0.86))$
 $= 756.444 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 249.647(1.0 \cdot 7.0 - 1.0 \cdot 2.989) - 2 \cdot 11.702$
 $(1.0 \cdot 7.0 - 1.0 \cdot 2.9894) \cdot (1 - 0.855)$



= 987.620 mm²

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot f_{r2}$$

$$= (2 \cdot 17.5) \cdot (11.7 - 0.46) \cdot 0.8551$$

$$= 336.422 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - A_r \text{ Lost}) \cdot f_{r3} + ((W_i - c_{an}/0.707)^2 - A_r \text{ Lost}) \cdot f_{r2} + W_p^2 \cdot f_{r4}$$

$$= (93.75^2) \cdot 0.86 + (0.0^2) \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 80.164 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (499.2934 - 273.05) \cdot 10.0 \cdot 1.0$$

$$= 2262.434 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 1.9606 \text{ mm}$
Wall Thickness per UG16(b),	$t_{r16b} = 5.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 5.9894 \text{ mm}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 5.9894 \text{ mm}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 5.5000 \text{ mm}$
Wall Thickness per table UG-45	$t_{b3} = 9.6026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[9.603, \max(5.9894, 5.5)]$$

$$= 5.9894 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.9606, 5.9894)$$

$$= 5.9894 \text{ mm}$$

Available Nozzle Neck Thickness = 13.2016 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 22.1,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 272.9 MPa	Passed
Occasional	: 2.0,	Allowable	: 156.9 MPa	Passed
Shear	: 14.0,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 13.202$, $t_r = 0.461$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.039$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 10.0$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.427$, Temp. Reduction = 44 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 10.0$, $c = 3.0 \text{ mm}$, $E^* = 1.0$



Thickness Ratio = $tr * E^* / (tg - c) = 0.427$, Temp. Reduction = 44 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 10.0$, $tr = 2.989$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.427$, Temp. Reduction = 44 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 10.0$, $tr = 2.989$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.427$, Temp. Reduction = 44 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.40 / 1.96 = 0.203$

Weld Size Calculations, Description: N110"

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 7.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$3.5000 = 0.5 * \text{TminPad}$	$7.0700 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (756.4436 - 987.6202 + 2 \cdot 11.7016 \cdot 0.8551 \cdot (1.0 \cdot 7.0 - 2.9894)) \cdot 138)$
 $= \max(0, -20824.52) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (336.4221 + 2262.4338 + 80.1645 - 0.0 \cdot 0.86) \cdot 138$
 $= 369666.72 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (336.4221 + 0.0 + 85.5088 + (140.0831)) \cdot 138$
 $= 77549.94 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (336.4221 + 0.0 + 80.1645 + 2262.4338 + (140.0831)) \cdot 138$
 $= 388996.19 \text{ N}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 273.05 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 247972. N$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 573.05 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 608615. N$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 130.6742) \cdot (13.2016 - 1.5) \cdot 0.7 \cdot 118$
 $= 396762. N$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 273.05 \cdot 10.0 \cdot 0.74 \cdot 138$
 $= 437961. N$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 273.05 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 262772. N$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (608615 + 396762) = 1005377 N$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (247972 + 437961 + 262772 + 0) = 948706 N$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (608615 + 262772 + 0) = 871387 N$

Summary of Failure Path Calculations:

Path 1-1 = 1005377 N	, must exceed W = 0 N	or W1 = 369666 N
Path 2-2 = 948705 N	, must exceed W = 0 N	or W2 = 77549 N
Path 3-3 = 871387 N	, must exceed W = 0 N	or W3 = 388996 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 9.0882 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 269.0882 mm

Input Echo, WRC297 Item 1, Description: N110" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2060.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle OD



Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	273.050	mm
Nozzle Thickness	13.2016	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	573.050	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	0.40	MPa
Radial Load	P	-6707.32	N
Circumferential Shear	Vc	10002.08	N
Longitudinal Shear	Vl	10002.08	N
Circumferential Moment	Mc	0.750E+07	N-mm
Longitudinal Moment	Ml	0.100E+08	N-mm
Torsional Moment	Mt	0.125E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(1041.5(10.0 - 3.0))}$$

$$= 554.819 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2083.000	mm
Nozzle Outside Diameter	(d) =	273.050	mm
Vessel Thickness used	(T) =	17.000	mm
Nozzle Thickness used	(t) =	11.702	mm
T / t	=	1.453	
d / t	=	23.334	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.451$$

Nr/P	=	0.076
Mr/P	=	0.130
M0/P	=	0.056
N0/P	=	0.140
MrD/Mc	=	0.239
NrDL/Mc	=	0.114
M0d/Mc	=	0.118
N0DL/Mc	=	0.114
MrD/Ml	=	0.150
NrDL/Ml	=	0.082
M0D/Ml	=	0.071
N0DL/Ml	=	0.216

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0



Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-27	-27	27	27
Outplane Bending (ML)	-53	53	53	-53
Normal Pressure Stress	24	24	24	24

Outplane Stress Summary	-46	46	116	-6
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-114	114	114	-114
Inplane Pressure Stress	12	12	12	12

Inplane Stress Summary	-92	99	156	-108
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6

Shear Stress Summary	7	7	4	4
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	94	100	157	108
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-67	67	67	-67
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	12	12	12	12

Outplane Stress Summary	-54	63	101	-48
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-136	136	136	-136
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	24	24	24	24

Inplane Stress Summary	-103	133	190	-117
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		6	6	6	6
Shear Stress Summary		4	4	7	7

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		103	133	191	118

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the hoop direction)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)		3	3	3	3
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		-27	-27	27	27
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		4	5	4	5
Hoop Stress Summary		-19	-18	35	36

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)		0	0	0	0
Axial Bending (P)		26	-26	26	-26
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		-16	-16	16	16
Axial Bending (ML)		-175	175	175	-175
Axial Pressure Stress		2	2	2	2
Axial Stress Summary		-162	135	222	-182

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress)		Outside	Inside	Outside	Inside
Shear due to (Vc)		1	1	-1	-1
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		9	9	9	9
Shear Stress Summary		11	11	7	7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Outside	Inside	Outside	Inside
Two * Max Shear Stress		162	155	222	219

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses in the hoop direction)		Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)		3	3	3	3



Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5

Hoop Stress Summary	-2	-2	18	19
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	26	-26	26	-26
Axial Membrane (Mc)	-12	-12	12	12
Axial Bending (Mc)	-218	218	218	-218
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-201	182	261	-230
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	9	9	9	9

Shear Stress Summary	7	7	11	11
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	201	185	261	250
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{1036.5(10.0 - 3.0)}$$

$$= 554.141 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2073.000 mm
Nozzle Outside Diameter	(d) =	573.050 mm
Vessel Thickness used	(T) =	7.000 mm
Nozzle Thickness used	(t) =	161.702 mm
T / t	=	0.043
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.757$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.048
Mr/P	=	0.072
M0/P	=	0.019
N0/P	=	0.022
MrD/Mc	=	0.241
NrDL/Mc	=	0.105
M0d/Mc	=	0.072
N0DL/Mc	=	0.044
MrD/Ml	=	0.049



NrDL/M1 = 0.054
 MØD/M1 = 0.015
 NØDL/M1 = 0.023

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-32	32	32	-32
Normal Pressure Stress	58	59	58	59
Outplane Stress Summary	37	69	117	22

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-19	-19	19	19
Inplane Bending (ML)	-105	105	105	-105
Inplane Pressure Stress	29	29	29	29
Inplane Stress Summary	-29	62	219	-109

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	67	72	219	131

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-115	115	115	-115
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29
Outplane Stress Summary	-79	119	175	-87

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to)	Cu Left	Cl Left	Du Right	Dl Right
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circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	-28	-28	28	28
Inplane Bending (Mc)	-386	386	386	-386
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	59	58	59
Inplane Stress Summary	-290	365	539	-352

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	5	5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	290	365	539	352

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	24	24	24	24	24	24	24	24
Circ. Pl (SUS)	-24	-24	30	30	-9	-9	11	11
Circ. Q (SUS)	-46	46	60	-60	-118	118	154	-154
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-9	-9	11	11	-7	-7	13	13
Long. Q (SUS)	-96	96	132	-132	-60	60	74	-74
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	24.0	24.0	24.0	24.0	24.0	24.0	24.0	24.0
Pm+Pl (SUS)	3.6	3.6	54.0	54.0	15.1	15.1	35.1	35.1
Pm+Pl+Q (Total)	94.0	99.9	155.6	109.2	103.5	133.4	189.5	119.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	138.00	Passed
Pm+Pl (SUS)	54.03	207.00	Passed
Pm+Pl+Q (TOTAL)	189.54	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	59	58	59	58	59	58	59
Circ. Pl (SUS)	-6	-6	10	10	-22	-22	34	34
Circ. Q (SUS)	-17	17	47	-47	-328	328	444	-444
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-13	-13	25	25	-9	-9	13	13
Long. Q (SUS)	-47	47	163	-163	-100	100	130	-130
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	58.0	59.0	58.0	59.0	58.0	59.0	58.0	59.0
Pm+Pl (SUS)	52.0	53.0	68.1	69.1	36.1	37.1	92.0	93.0
Pm+Pl+Q (Total)	66.5	71.8	217.0	131.1	292.0	365.0	536.0	351.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	59.00	138.00	Passed
Pm+Pl (SUS)	93.02	207.00	Passed
Pm+Pl+Q (TOTAL)	536.04	413.99	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	5	4	5	4	5	4	5
Circ. Pl (SUS)	-24	-24	30	30	-7	-7	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-16	-16	16	16	-12	-12	12	12
Long. Q (SUS)	-149	149	201	-201	-192	192	244	-244
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	4.0	5.0	4.0	5.0	4.0	5.0	4.0	5.0
Pm+Pl (SUS)	20.2	19.2	34.1	35.1	10.1	10.1	17.3	18.2
Pm+Pl+Q (Total)	163.7	155.3	219.3	218.6	202.3	184.7	258.4	248.8

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	118.00	Passed
Pm+Pl (SUS)	35.06	177.00	Passed
Pm+Pl+Q (TOTAL)	258.41	354.00	Passed



*Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.*

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Input, Nozzle Desc: N24"

Parent: N110"

Pressure for Reinforcement Calculations	P	0.3987	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Diameter of Cylindrical Shell	D	246.65	mm
Shell Finished (Minimum) Thickness	t	13.2016	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	6.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	216.0000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		50.8500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500	in.
Actual Thickness Used in Calculation	0.207	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 0.4 \cdot 124.8233 / (118 \cdot 1.0 - 0.6 \cdot 0.4)$$

$$= 0.4226 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 0.4 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.4)$$

$$= 0.1928 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	213.5307	mm
Parallel to Vessel Wall, opening length	d	106.7653	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.4183	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 118.0 / 118.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 118.0 / 118.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr4]:

$$= \min(1, S_p / S_v)$$

$$= \min(1, 138.0 / 118.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1.0, 1.0)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	45.120	NA	NA
Area in Shell	A1	1204.211	NA	NA
Area in Nozzle Wall	A2	181.715	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	36.000	NA	NA
Area in Element	A5	1587.691	NA	NA
TOTAL AREA AVAILABLE	Atot	3009.617	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (106.7653 \cdot 0.4226 \cdot 1.0 + 2 \cdot 3.7673 \cdot 0.4226 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 45.120 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 106.765(1.0 \cdot 11.7016 - 1.0 \cdot 0.423) - 2 \cdot 3.767$$

$$(1.0 \cdot 11.7016 - 1.0 \cdot 0.4226) \cdot (1 - 1.0)$$

$$= 1204.211 \text{ mm}^2$$



Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lp}) \cdot (t_n - t_{rn}) \cdot f_{r2}$$

$$= (2 \cdot 25.42) \cdot (3.77 - 0.19) \cdot 1.0$$

$$= 181.715 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot f_{r3} + (W_i - c_{an}/0.707)^2 \cdot f_{r2} + W_p^2 \cdot f_{r4}$$

$$= 6.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$$

$$= 36.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot f_{r4}$$

$$= (213.5307 - 114.3) \cdot 16.0 \cdot 1.0$$

$$= 1587.691 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 1.6928 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 1.9226 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 4.0000 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.0000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 6.7578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[6.758, \max(4.0, 4.0)]$$

$$= 4.0000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.6928, 4.0)$$

$$= 4.0000 \text{ mm}$$

Available Nozzle Neck Thickness = 5.2673 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	58.1,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	236.9 MPa	Passed
Occasional	:	2.7,	Allowable	:	156.9 MPa	Passed
Shear	:	38.6,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } t_g = 5.267, tr = 0.193, c = 1.5 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.051, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 5.267, tr = 0.193, c = 1.5 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.051, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 5.267, tr = 0.193, c = 1.5 \text{ mm, } E^* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.051, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 13.202, tr = 0.423, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.036$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 5.267, tr = 0.193, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.051$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.40 / 5.10 = 0.078$

Weld Size Calculations, Description: N24"

Intermediate Calc. for nozzle/shell Welds Tmin 3.7673 mm
 Intermediate Calc. for pad/shell Welds TminPad 11.7016 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.6371 = 0.7 \cdot t_{min}$	$4.2420 = 0.7 \cdot W_o$ mm
Pad Weld	$5.8508 = 0.5 \cdot T_{minPad}$	$8.4840 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (45.1196 - 1204.2111 + 2 \cdot 3.7673 \cdot 1.0 \cdot (1.0 \cdot 11.7016 - 0.4226)) \cdot 118)$
 $= \max(0, -126733.94) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (181.715 + 1587.691 + 36.0 - 0.0 \cdot 1.0) \cdot 118$
 $= 213019.78 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (181.715 + 0.0 + 36.0 + (88.1678)) \cdot 118$
 $= 36091.10 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (181.715 + 0.0 + 36.0 + 1587.691 + (88.1678)) \cdot 118$
 $= 223422.69 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 6.0 \cdot 0.49 \cdot 118$$

$$= 62281. N$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 216.0 \cdot 12.0 \cdot 0.49 \cdot 118$$

$$= 235394. N$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 55.2663) \cdot (5.2673 - 1.5) \cdot 0.7 \cdot 118$$

$$= 54024. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 114.3 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 293327. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (13.2016 - 1.5) \cdot 0.74 \cdot 118$$

$$= 183438. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (235394 + 54024) = 289418 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (62281 + 293327 + 183438 + 0) = 539047 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (235394 + 183438 + 0) = 418833 N$$

Summary of Failure Path Calculations:

Path 1-1 = 289418 N , must exceed W = 0 N	or W1 = 213019 N
Path 2-2 = 539046 N , must exceed W = 0 N	or W2 = 36091 N
Path 3-3 = 418832 N , must exceed W = 0 N	or W3 = 223422 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 14.0415 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 277.2432 mm

Input Echo, WRC297 Item 1, Description: N24" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	246.647	mm
Shell Thickness	13.2016	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-106 B	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	114.300	mm



Nozzle Thickness 5.2673 mm
 Nozzle Stress Concentration Factor 1.000
 Nozzle Material SA-106 B
 Nozzle Cold S.I. Allowable SNmc 118.00 MPa
 Nozzle Hot S.I. Allowable SNmh 118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad 216.000 mm
 Thickness of Reinforcing Pad 16.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure Dp 0.40 MPa
 Radial Load P -2706.46 N
 Circumferential Shear Vc 4000.85 N
 Longitudinal Shear Vl 4000.85 N
 Circumferential Moment Mc 0.121E+07 N-mm
 Longitudinal Moment Ml 0.161E+07 N-mm
 Torsional Moment Mt 0.200E+07 N-mm

Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(138.674(13.202 - 1.5))}$
 $= 247.234 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 277.348 mm
 Nozzle Outside Diameter (d) = 114.300 mm
 Vessel Thickness used (T) = 27.702 mm
 Nozzle Thickness used (t) = 3.767 mm
 T / t = 7.353
 d / t = 30.340

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.304$$

Nr/P = 0.056
 Mr/P = 0.036
 M0/P = 0.169
 N0/P = 0.366
 MrD/Mc = 0.064
 NrDL/Mc = 0.078
 M0d/Mc = 0.309
 N0DL/Mc = 0.245
 MrD/Ml = 0.059
 NrDL/Ml = 0.073
 M0D/Ml = 0.262
 N0DL/Ml = 0.393

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7



Outplane Bending (ML)	-28	28	28	-28
Normal Pressure Stress	1	2	1	2

Outplane Stress Summary	-29	21	42	-21
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Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	0	0	0	0
Inplane Stress Summary	-6	5	9	-4

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	30	22	42	22

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-25	25	25	-25
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	0	0	0	0
Outplane Stress Summary	-23	20	34	-23

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-5	5	5	-5
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1	2	1	2
Inplane Stress Summary	-3	5	8	-2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	23	21	35	24

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	0	1	15	15

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	9	-9	9	-9
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-45	-45	45	45
Axial Bending (ML)	-134	134	134	-134
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-165	83	195	-93

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	5	5	-5	-5
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	25	25	25	25
Shear Stress Summary	31	31	19	19

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	177	104	197	115

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7

Hoop Stress Summary	4	4	11	11
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	2	2	2	2
Axial Bending (P)	9	-9	9	-9
Axial Membrane (Mc)	-34	-34	34	34
Axial Bending (Mc)	-109	109	109	-109
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-128	70	159	-79
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	25	25	25	25

Shear Stress Summary	19	19	31	31
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	139	76	166	111
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{130.674(13.202 - 1.5)}$
 $= 243.342 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	261.348 mm
Nozzle Outside Diameter (d)	=	216.000 mm
Vessel Thickness used (T)	=	11.702 mm
Nozzle Thickness used (t)	=	54.617 mm
T / t	=	0.214
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.906$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.050
Mr/P	=	0.084
M0/P	=	0.026
N0/P	=	0.055
MrD/Mc	=	0.256
NrDL/Mc	=	0.090
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/M1	=	0.069
NrDL/M1	=	0.049
M0D/M1	=	0.021



NØDL/M1 = 0.059

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	4	4	4	4
Outplane Stress Summary	-1	6	18	-1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	9	-9	9	-9
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	2	2	2	2
Inplane Stress Summary	-12	12	38	-26

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	13	14	38	26

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	2	2	2	2
Outplane Stress Summary	-15	15	28	-15

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	9	-9	9	-9
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-62	62	62	-62
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	4	4	4	4

Inplane Stress Summary	-51	54	81	-63
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	51	54	81	63

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	2	1	2	1	2	1	2
Circ. Pl (SUS)	-6	-6	8	8	-1	-1	1	1
Circ. Q (SUS)	-25	25	31	-31	-5	5	5	-5
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-1	-1	1	1	-2	-2	4	4
Long. Q (SUS)	-6	6	6	-6	-22	22	28	-28
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0
Pm+Pl (SUS)	5.0	4.0	9.0	10.0	2.0	3.0	4.0	4.0
Pm+Pl+Q (Total)	30.4	21.5	40.3	21.5	24.5	20.6	32.4	24.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	118.00	Passed
Pm+Pl (SUS)	10.00	177.00	Passed
Pm+Pl+Q (TOTAL)	40.27	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	4	4	4	4	4	4	4
Circ. Pl (SUS)	-2	-2	4	4	-3	-3	3	3
Circ. Q (SUS)	-3	3	9	-9	-53	53	71	-71
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-2	-2	2	2	-2	-2	4	4
Long. Q (SUS)	-13	13	31	-31	-15	15	21	-21
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Pm+Pl (SUS)	2.8	2.8	8.2	8.2	2.2	2.2	7.6	7.6
Pm+Pl+Q (Total)	13.7	14.0	35.1	27.0	52.0	54.0	78.2	64.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	118.00	Passed
Pm+Pl (SUS)	8.24	177.00	Passed
Pm+Pl+Q (TOTAL)	78.18	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. Pl (SUS)	-6	-6	8	8	-2	-2	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-43	-43	47	47	-32	-32	36	36
Long. Q (SUS)	-125	125	143	-143	-100	100	118	-118
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	25	25	25	25	25	25	25	25
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+Pl (SUS)	41.2	42.2	50.7	50.7	34.5	35.4	39.8	39.9
Pm+Pl+Q (Total)	175.6	103.2	195.2	115.2	138.9	77.2	162.9	108.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed
Pm+Pl (SUS)	50.70	177.00	Passed
Pm+Pl+Q (TOTAL)	195.21	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.





Input, Nozzle Desc: N54"

From: 40

Pressure for Reinforcement Calculations	P	0.3800	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2060.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4330.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	230.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	6.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	203.2000	mm
Thickness of Pad	te	6.0000	mm
Weld leg size between Pad and Shell	Wp	6.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	6.0000	mm
Reinforcing Pad Width		44.4500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N54"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500	in.
Actual Thickness Used in Calculation	0.295	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.38 \cdot 1033.0 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$
 $= 2.8494 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.38 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.38)$
 $= 0.1838 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	204.6407	mm
Parallel to Vessel Wall, opening length	d	102.3204	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		17.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	296.499	NA	NA
Area in Shell	A1	417.485	NA	NA
Area in Nozzle Wall	A2	173.763	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	38.909	NA	NA
Area in Element	A5	533.400	NA	NA
TOTAL AREA AVAILABLE	Atot	1163.556	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (102.3204 \cdot 2.8494 \cdot 1.0 + 2 \cdot 5.9898 \cdot 2.8494 \cdot 1.0 \cdot (1 - 0.86))$
 $= 296.499 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 102.32(1.0 \cdot 7.0 - 1.0 \cdot 2.849) - 2 \cdot 5.99$
 $(1.0 \cdot 7.0 - 1.0 \cdot 2.8494) \cdot (1 - 0.855)$
 $= 417.485 \text{ mm}^2$



Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 17.5) \cdot (5.99 - 0.18) \cdot 0.8551$$

$$= 173.763 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$$

$$= (36.0) \cdot 0.86 + (0.0) \cdot 0.86 + 8.1254^2 \cdot 1.0$$

$$= 38.909 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (203.2 - 114.3) \cdot 6.0 \cdot 1.0$$

$$= 533.400 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.6838 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 5.8494 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.8494 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 6.7578 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.758, \max(5.8494, 5.5)]$$

$$= 5.8494 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.6838, 5.8494)$$

$$= 5.8494 \text{ mm}$$

Available Nozzle Neck Thickness = 7.4898 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 38.5,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 256.5 MPa	Passed
Occasional	: 1.5,	Allowable	: 156.9 MPa	Passed
Shear	: 26.2,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.49, tr = 0.184, c = 1.5 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.031, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 6.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.407, Temp. Reduction = 49 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 6.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.407, Temp. Reduction = 49 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 6.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.407$, Temp. Reduction = 49 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.49, tr = 0.184, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.031$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.38/5.10 = 0.075$

Weld Size Calculations, Description: N54"

Intermediate Calc. for nozzle/shell Welds Tmin 5.9898 mm
 Intermediate Calc. for pad/shell Welds TminPad 6.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.1929 = 0.7 \cdot t_{min}$	$4.2420 = 0.7 \cdot W_o$ mm
Pad Weld	$3.0000 = 0.5 \cdot t_{minPad}$	$4.2420 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (296.499 - 417.4845 + 2 \cdot 5.9898 \cdot 0.8551 \cdot (1.0 \cdot 7.0 - 2.8494)) \cdot 138)$
 $= \max(0, -10827.51) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (173.7627 + 533.4 + 38.9085 - 0.0 \cdot 0.86) \cdot 138$
 $= 102947.21 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (173.7627 + 0.0 + 30.7832 + (71.7056)) \cdot 138$
 $= 38118.77 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (173.7627 + 0.0 + 38.9085 + 533.4 + (71.7056)) \cdot 138$
 $= 112841.55 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 114.3 \cdot 6.0 \cdot 0.49 \cdot 118$
 $= 62281. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 203.2 \cdot 6.0 \cdot 0.49 \cdot 138$
 $= 129487. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 54.1551) \cdot (7.4898 - 1.5) \cdot 0.7 \cdot 118$
 $= 84168. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 114.3 \cdot 6.0 \cdot 0.74 \cdot 138$
 $= 110000. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 114.3 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 109998. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (129487 + 84168) = 213655 \text{ N}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (62281 + 110000 + 109998 + 0) = 282279 \text{ N}$

Failure Path [Path 3-3]:
 $= (S_{pew} + T_{ngw} + S_{inw})$
 $= (129487 + 109998 + 0) = 239484 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 213654 N	, must exceed W = 0 N	or W1 = 102947 N
Path 2-2 = 282278 N	, must exceed W = 0 N	or W2 = 38118 N
Path 3-3 = 239484 N	, must exceed W = 0 N	or W3 = 112841 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5867 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.5867 mm

Input Echo, WRC297 Item 1, Description: N54" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2060.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	7.4898	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	203.200	mm
Thickness of Reinforcing Pad	6.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	0.38	MPa
Radial Load	P	-2706.46	N
Circumferential Shear	Vc	4000.85	N
Longitudinal Shear	Vl	4000.85	N
Circumferential Moment	Mc	0.121E+07	N-mm
Longitudinal Moment	Ml	0.161E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	Yes
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Indices for Nozzles in Cylindrical Shells per Table 5-D.2

Stress	Longitudinal Plane		Transverse Plane	
	Inside Corner	Outside Corner	Inside Corner	Outside Corner
sn	3.100	1.200	1.000	2.100
st	-0.200	1.000	-0.200	2.600

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1039.5(10.0 - 3.0))}$$

$$= 395.798 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2079.000 mm
Nozzle Outside Diameter	(d) =	114.300 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	5.990 mm
T / t	=	2.170
d / t	=	19.082

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.695$$

Nr/P	=	0.130
Mr/P	=	0.157
M0/P	=	0.105
N0/P	=	0.206
MrD/Mc	=	0.217
NrDL/Mc	=	0.157
M0d/Mc	=	0.170
N0DL/Mc	=	0.124
MrD/Ml	=	0.190
NrDL/Ml	=	0.143
M0D/Ml	=	0.143
N0DL/Ml	=	0.139



Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-11	-11	11	11
Outplane Bending (ML)	-71	71	71	-71
Normal Pressure Stress	36	94	36	94
Outplane Stress Summary	-33	147	133	27

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-94	94	94	-94
Inplane Pressure Stress	30	-6	30	-6
Inplane Stress Summary	-59	63	154	-101

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	7	7	7	7
Shear Stress Summary	9	9	5	5

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	62	148	155	129

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-63	63	63	-63
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	30	63	30
Outplane Stress Summary	5	79	148	-32

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2



Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-81	81	81	-81
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	79	-6	79	-6

Inplane Stress Summary	4	52	187	-90
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	7	7	7	7

Shear Stress Summary	5	5	9	9
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	11	80	189	92
------------------------	----	----	-----	----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4

Hoop Stress Summary	-4	-4	18	18
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	41	-41	41	-41
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-30	-30	30	30
Axial Bending (ML)	-278	278	278	-278
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-264	209	353	-286
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	16	16	16	16

Shear Stress Summary	19	19	12	12
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside



Two * Max Shear Stress 265 217 354 306

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses in the
 hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4

Hoop Stress Summary 0 0 14 15

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	41	-41	41	-41
Axial Membrane (Mc)	-22	-22	22	22
Axial Bending (Mc)	-245	245	245	-245
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary -223 183 312 -260

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	16	16	16	16

Shear Stress Summary 12 12 19 19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress 224 185 314 278

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{1036.5(10.0 - 3.0)}$
 $= 395.391 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2073.000 mm
Nozzle Outside Diameter	(d) =	203.200 mm
Vessel Thickness used	(T) =	7.000 mm
Nozzle Thickness used	(t) =	50.440 mm
T / t	=	0.139
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.687$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.



Nr/P = 0.089
 Mr/P = 0.165
 M0/P = 0.048
 N0/P = 0.070
 MrD/Mc = 0.296
 NrDL/Mc = 0.100
 M0d/Mc = 0.091
 N0DL/Mc = 0.062
 MrD/Ml = 0.128
 NrDL/Ml = 0.110
 M0D/Ml = 0.039
 N0DL/Ml = 0.089

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-37	37	37	-37
Normal Pressure Stress	55	56	55	56
Outplane Stress Summary	23	67	127	20

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-17	-17	17	17
Inplane Bending (ML)	-124	124	124	-124
Inplane Pressure Stress	27	27	27	27
Inplane Stress Summary	-54	84	229	-128

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	78	86	229	148

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-66	66	66	-66



Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	27	27	27	27
Outplane Stress Summary	-25	74	121	-42

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-215	215	215	-215
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	55	56	55	56
Inplane Stress Summary	-112	209	343	-196

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	112	209	343	197

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	36	94	36	94	79	-6	79	-6
Circ. Pl (SUS)	-8	-8	14	14	-7	-7	11	11
Circ. Q (SUS)	-61	61	81	-81	-66	66	96	-96
Long. Pm (SUS)	30	-6	30	-6	63	30	63	30
Long. Pl (SUS)	-9	-9	13	13	-4	-4	10	10
Long. Q (SUS)	-79	79	109	-109	-53	53	73	-73
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	36.0	100.0	36.0	100.0	79.0	36.0	79.0	36.0
Pm+Pl (SUS)	28.1	101.0	50.1	108.0	72.1	39.1	90.1	40.0
Pm+Pl+Q (Total)	60.3	147.8	153.6	129.6	12.0	80.3	187.5	92.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	100.00	138.00	Passed
Pm+Pl (SUS)	108.01	207.00	Passed
Pm+Pl+Q (TOTAL)	187.54	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	55	56	55	56	55	56	55	56
Circ. Pl (SUS)	-11	-11	17	17	-8	-8	16	16
Circ. Q (SUS)	-22	22	52	-52	-161	161	269	-269
Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-13	-13	21	21	-4	-4	10	10
Long. Q (SUS)	-70	70	178	-178	-51	51	81	-81
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	55.0	56.0	55.0	56.0	55.0	56.0	55.0	56.0
Pm+Pl (SUS)	44.0	45.0	72.0	73.0	47.0	48.0	71.0	72.0
Pm+Pl+Q (Total)	78.6	85.4	226.1	151.1	114.1	209.1	340.1	197.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	56.00	138.00	Passed
Pm+Pl (SUS)	73.04	207.00	Passed
Pm+Pl+Q (TOTAL)	340.11	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	4	3	4	3	4	3	4
Circ. Pl (SUS)	-8	-8	14	14	-4	-4	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-29	-29	31	31	-21	-21	23	23
Long. Q (SUS)	-237	237	319	-319	-204	204	286	-286
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	16	16	16	16	16	16	16	16
Pm (SUS)	3.0	4.0	3.0	4.0	3.0	4.0	3.0	4.0
Pm+Pl (SUS)	28.4	28.4	32.6	32.6	20.5	20.9	24.8	24.8
Pm+Pl+Q (Total)	266.4	216.4	351.5	306.1	224.8	185.8	311.2	278.6



Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	118.00	Passed
Pm+Pl (SUS)	32.62	177.00	Passed
Pm+Pl+Q (TOTAL)	351.51	354.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N64"

From: 40

Pressure for Reinforcement Calculations	P	0.3800	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	2060.00	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4930.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	230.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	6.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	203.2000	mm
Thickness of Pad	te	6.0000	mm
Weld leg size between Pad and Shell	Wp	6.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	6.0000	mm
Reinforcing Pad Width		44.4500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N64"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500	in.
Actual Thickness Used in Calculation	0.383	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.38 \cdot 1033.0 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$
 $= 2.8494 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.38 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.38)$
 $= 0.1838 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	195.6618	mm
Parallel to Vessel Wall, opening length	d	97.8309	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		17.5000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	285.560	NA	NA
Area in Shell	A1	396.150	NA	NA
Area in Nozzle Wall	A2	240.943	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	30.783	NA	NA
Area in Element	A5	488.171	NA	NA
TOTAL AREA AVAILABLE	Atot	1156.047	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (97.8309 \cdot 2.8494 \cdot 1.0 + 2 \cdot 8.2345 \cdot 2.8494 \cdot 1.0 \cdot (1 - 0.86))$
 $= 285.560 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 97.831(1.0 \cdot 7.0 - 1.0 \cdot 2.849) - 2 \cdot 8.235$
 $(1.0 \cdot 7.0 - 1.0 \cdot 2.8494) \cdot (1 - 0.855)$

= 396.150 mm²

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot f_{r2}$$

$$= (2 \cdot 17.5) \cdot (8.23 - 0.18) \cdot 0.8551$$

$$= 240.943 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot f_{r3} + (W_i - c_{an}/0.707)^2 \cdot f_{r2} + W_p^2 \cdot f_{r4}$$

$$= 6.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 30.783 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (195.6618 - 114.3) \cdot 6.0 \cdot 1.0$$

$$= 488.171 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 1.6838 \text{ mm}$
Wall Thickness per UG16(b),	$t_{r16b} = 5.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 5.8494 \text{ mm}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 5.8494 \text{ mm}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 5.5000 \text{ mm}$
Wall Thickness per table UG-45	$t_{b3} = 6.7578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[6.7578, \max(5.8494, 5.5)]$$

$$= 5.8494 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.6838, 5.8494)$$

$$= 5.8494 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29.7,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 265.3 MPa	Passed
Occasional	: 1.0,	Allowable	: 156.9 MPa	Passed
Shear	: 20.6,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.735$, $t_r = 0.184$, $c = 1.5 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 6.0$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.407$, Temp. Reduction = 49 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



Govrn. thk, tg = 6.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.407$, Temp. Reduction = 49 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 6.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.407$, Temp. Reduction = 49 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.735, tr = 0.184, c = 1.5 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.38/5.10 = 0.075$

Weld Size Calculations, Description: N64"

Intermediate Calc. for nozzle/shell Welds Tmin 6.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 6.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.2000 = 0.7 \cdot t_{min}$	$4.2420 = 0.7 \cdot W_o$ mm
Pad Weld	$3.0000 = 0.5 \cdot t_{minPad}$	$4.2420 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (285.5605 - 396.1504 + 2 \cdot 8.2345 \cdot 0.8551 \cdot (1.0 \cdot 7.0 - 2.8494)) \cdot 138)$
 $= \max(0, -7194.45) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot Sv$
 $= (240.943 + 488.1708 + 30.7832 - 0.0 \cdot 0.86) \cdot 138$
 $= 104854.96 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot Sv$
 $= (240.943 + 0.0 + 30.7832 + (98.5777)) \cdot 138$
 $= 51096.66 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (240.943 + 0.0 + 30.7832 + 488.1708 + (98.5777)) \cdot 138$
 $= 118457.28 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 6.0 \cdot 0.49 \cdot 118$$

$$= 62281. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 203.2 \cdot 6.0 \cdot 0.49 \cdot 138$$

$$= 129487. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 53.0327) \cdot (9.7345 - 1.5) \cdot 0.7 \cdot 118$$

$$= 113312. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 114.3 \cdot 6.0 \cdot 0.74 \cdot 138$$

$$= 110000. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 109998. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{SNW}) = (129487 + 113312) = 242799 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (62281 + 110000 + 109998 + 0) = 282279 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (129487 + 109998 + 0) = 239484 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 242798 N	, must exceed W = 0 N	or W1 = 104854 N
Path 2-2 = 282278 N	, must exceed W = 0 N	or W2 = 51096 N
Path 3-3 = 239484 N	, must exceed W = 0 N	or W3 = 118457 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.5867 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.5867 mm

Input Echo, WRC297 Item 1, Description: N64" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2060.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	203.200	mm
Thickness of Reinforcing Pad	6.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	0.38	MPa
Radial Load	P	-2706.46	N
Circumferential Shear	Vc	4000.85	N
Longitudinal Shear	Vl	4000.85	N
Circumferential Moment	Mc	0.121E+07	N-mm
Longitudinal Moment	Ml	0.161E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	Yes
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Indices for Nozzles in Cylindrical Shells per Table 5-D.2

Stress	Longitudinal Plane		Transverse Plane	
	Inside Corner	Outside Corner	Inside Corner	Outside Corner
sn	3.100	1.200	1.000	2.100
st	-0.200	1.000	-0.200	2.600

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1039.5(10.0 - 3.0))}$$

$$= 395.798 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2079.000 mm
Nozzle Outside Diameter	(d) =	114.300 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	8.235 mm
T / t	=	1.579
d / t	=	13.881

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.695$$

Nr/P	=	0.146
Mr/P	=	0.191
M0/P	=	0.090
N0/P	=	0.167
MrD/Mc	=	0.251
NrDL/Mc	=	0.160
M0d/Mc	=	0.142
N0DL/Mc	=	-0.079
MrD/Ml	=	0.211
NrDL/Ml	=	0.143



M0D/M1 = 0.118
 N0DL/M1 = 0.095

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-58	58	58	-58
Normal Pressure Stress	36	94	36	94
Outplane Stress Summary	-19	139	114	37

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-105	105	105	-105
Inplane Pressure Stress	30	-6	30	-6
Inplane Stress Summary	-66	71	168	-115

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	7	7	7	7
Shear Stress Summary	9	9	5	5

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	68	140	169	153

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	4	4	-4	-4
Outplane Bending (Mc)	-53	53	53	-53
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	30	63	30
Outplane Stress Summary	26	82	123	-33

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-93	93	93	-93
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	79	-6	79	-6

Inplane Stress Summary	-4	61	203	-105
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	7	7	7	7

Shear Stress Summary	5	5	9	9
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	32	84	204	107

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2

Hoop Stress Summary	-2	-2	13	13
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	28	-28	28	-28
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-23	-23	23	23
Axial Bending (ML)	-173	173	173	-173
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-166	123	227	-175
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11

Shear Stress Summary	14	14	9	9
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Nozzle Stresses (MPa):



LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	167	129	227	190

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	4	4	-4	-4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	10	10	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	28	-28	28	-28
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-159	159	159	-159
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-146	115	207	-167

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	11	11	11	11
Shear Stress Summary	9	9	14	14

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	157	116	209	170

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1036.5(10.0 - 3.0))}$
 $= 395.391 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2073.000 mm
Nozzle Outside Diameter	(d) =	203.200 mm
Vessel Thickness used	(T) =	7.000 mm
Nozzle Thickness used	(t) =	52.685 mm
T / t	=	0.133
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.687$$



Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.090
 Mr/P = 0.165
 M0/P = 0.048
 N0/P = 0.069
 MrD/Mc = 0.297
 NrDL/Mc = 0.102
 M0d/Mc = 0.091
 N0DL/Mc = 0.061
 MrD/Ml = 0.128
 NrDL/Ml = 0.112
 M0D/Ml = 0.039
 N0DL/Ml = 0.087

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-37	37	37	-37
Normal Pressure Stress	55	56	55	56
Outplane Stress Summary	23	68	127	20

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-18	-18	18	18
Inplane Bending (ML)	-124	124	124	-124
Inplane Pressure Stress	27	27	27	27
Inplane Stress Summary	-54	84	229	-127

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	79	86	229	148

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	15	-15	15	-15



Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-66	66	66	-66
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	27	27	27	27

Outplane Stress Summary	-25	74	121	-43
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-215	215	215	-215
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	55	56	55	56

Inplane Stress Summary	-112	209	343	-196
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	6	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	112	209	343	196
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	36	94	36	94	79	-6	79	-6
Circ. Pl (SUS)	-5	-5	9	9	-7	-7	11	11
Circ. Q (SUS)	-50	50	66	-66	-75	75	111	-111
Long. Pm (SUS)	30	-6	30	-6	63	30	63	30
Long. Pl (SUS)	-9	-9	13	13	6	6	-2	-2
Long. Q (SUS)	-87	87	123	-123	-45	45	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	36.0	100.0	36.0	100.0	79.0	36.0	79.0	36.0
Pm+Pl (SUS)	31.1	104.0	45.4	103.0	72.3	49.0	90.0	28.0
Pm+Pl+Q (Total)	67.3	139.9	166.6	153.5	29.5	82.7	201.8	106.9

Vessel Stress Summation Comparison (MPa):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	100.00	138.00	Passed
Pm+Pl (SUS)	104.02	207.00	Passed
Pm+Pl+Q (TOTAL)	201.80	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	55	56	55	56	55	56	55	56
Circ. Pl (SUS)	-11	-11	17	17	-8	-8	16	16
Circ. Q (SUS)	-22	22	52	-52	-161	161	269	-269
Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-14	-14	22	22	-4	-4	10	10
Long. Q (SUS)	-70	70	178	-178	-51	51	81	-81
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	55.0	56.0	55.0	56.0	55.0	56.0	55.0	56.0
Pm+Pl (SUS)	44.0	45.0	72.0	73.0	47.0	48.0	71.0	72.0
Pm+Pl+Q (Total)	79.6	84.4	227.1	150.1	114.1	209.1	340.1	197.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	56.00	138.00	Passed
Pm+Pl (SUS)	73.04	207.00	Passed
Pm+Pl+Q (TOTAL)	340.11	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	-5	-5	9	9	6	6	-2	-2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-23	-23	23	23	-17	-17	17	17
Long. Q (SUS)	-145	145	201	-201	-131	131	187	-187
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	22.2	22.2	24.3	24.3	24.3	24.3	18.4	18.4



Pm+Pl+Q (Total)	168.0	128.7	225.4	188.9	156.0	115.8	206.6	171.0
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Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	118.00	Passed
Pm+Pl (SUS)	24.33	177.00	Passed
Pm+Pl+Q (TOTAL)	225.38	354.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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